



2026년도 첨단소재 원천기술 성장지원사업(2026-나노소재-6차-지정공모형-01)

1차년도 Kick-off 워크숍

10 kg/h급 연속식 분말 ALD 공정을 이용한 나트륨이온전 지용 고전압 양극재 및 1Ah급 파우치셀 기술 개발

2026. 08. 26. 수요일 14:30-18:00 @ 무주 리조트

한국기계연구원 이승모

주관연구개발기관 전북대학교

박상호

공동연구개발기관 대성기계공업(주) (이용남)

공동연구개발기관 한국기계연구원 (이승모)

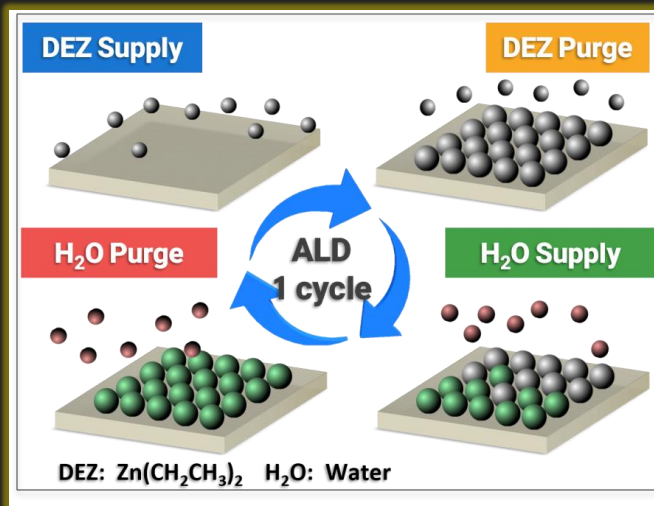
공동연구개발기관 (주)씨오알엔 (성동길)

공동연구개발기관 (주)에버인더스(박지훈)

목 차

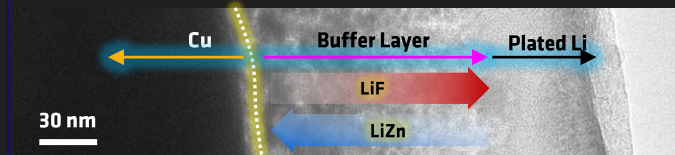
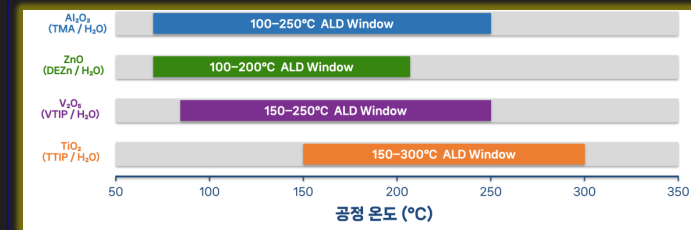
화제 1.

원자층 단위 증착 공정(ALD)



화제 2.

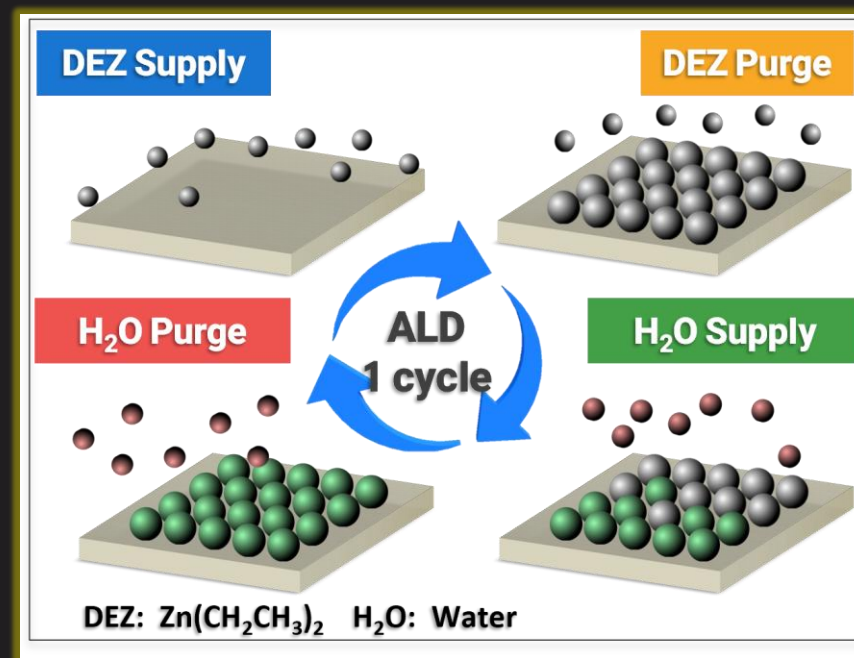
1-3차년도 연구 계획



목 차

화제 1.

원자층 단위 증착 공정(ALD)



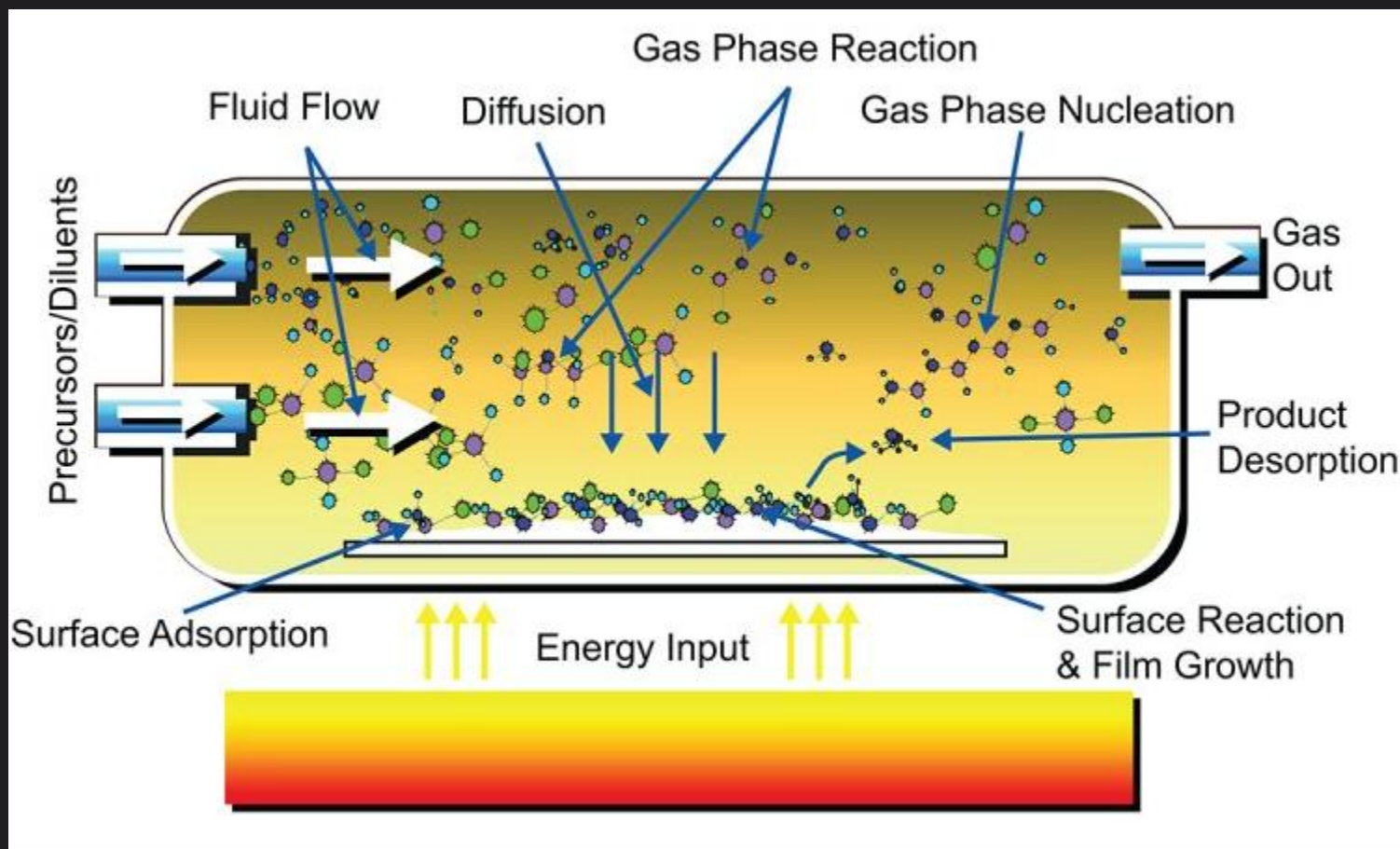
Atomic Layer Deposition (ALD)



Atomic Layer Deposition (ALD)

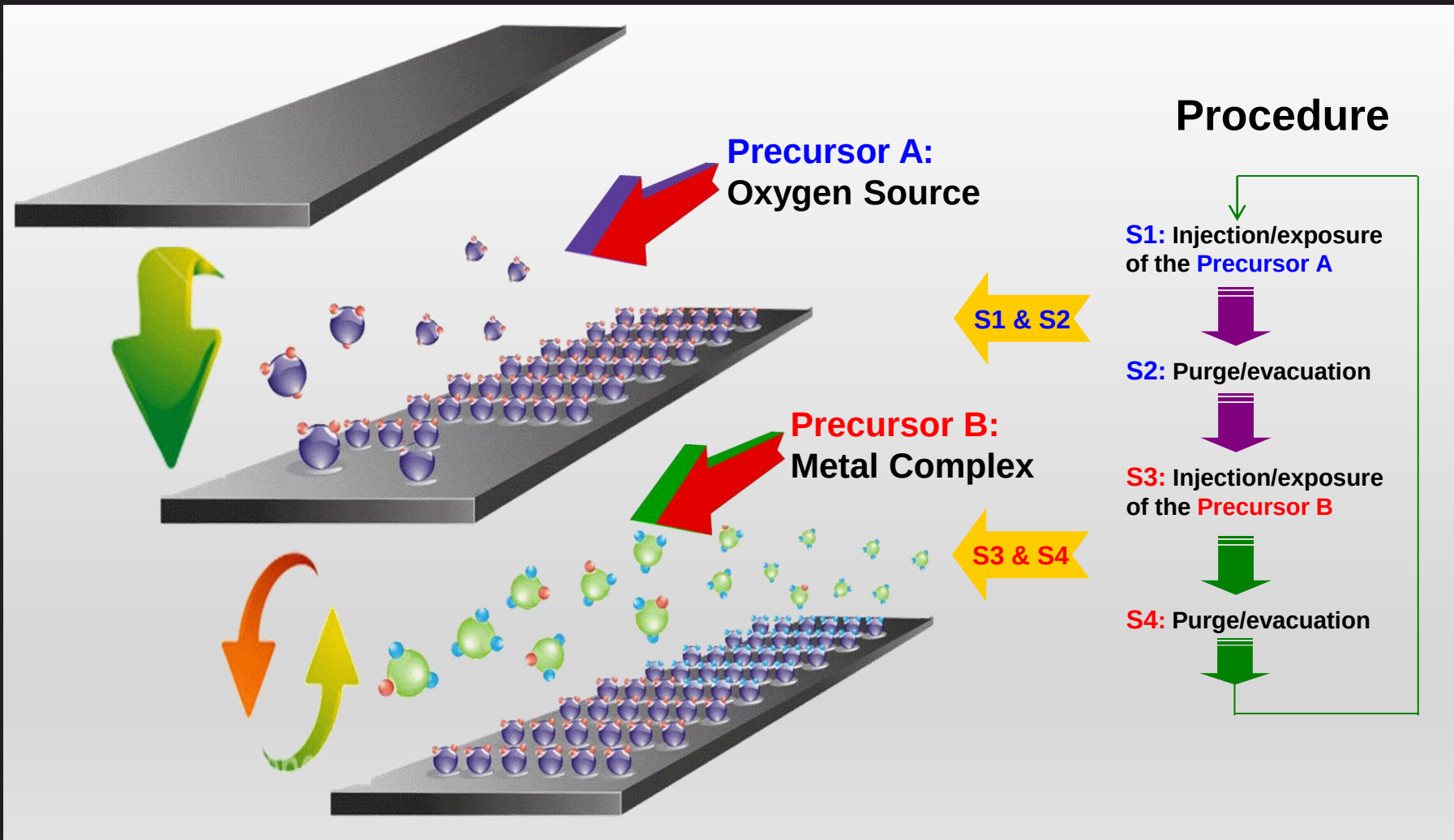


ALD ???



What if we split the CVD reaction into two half-reactions?

ALD ???





Principle of ALD

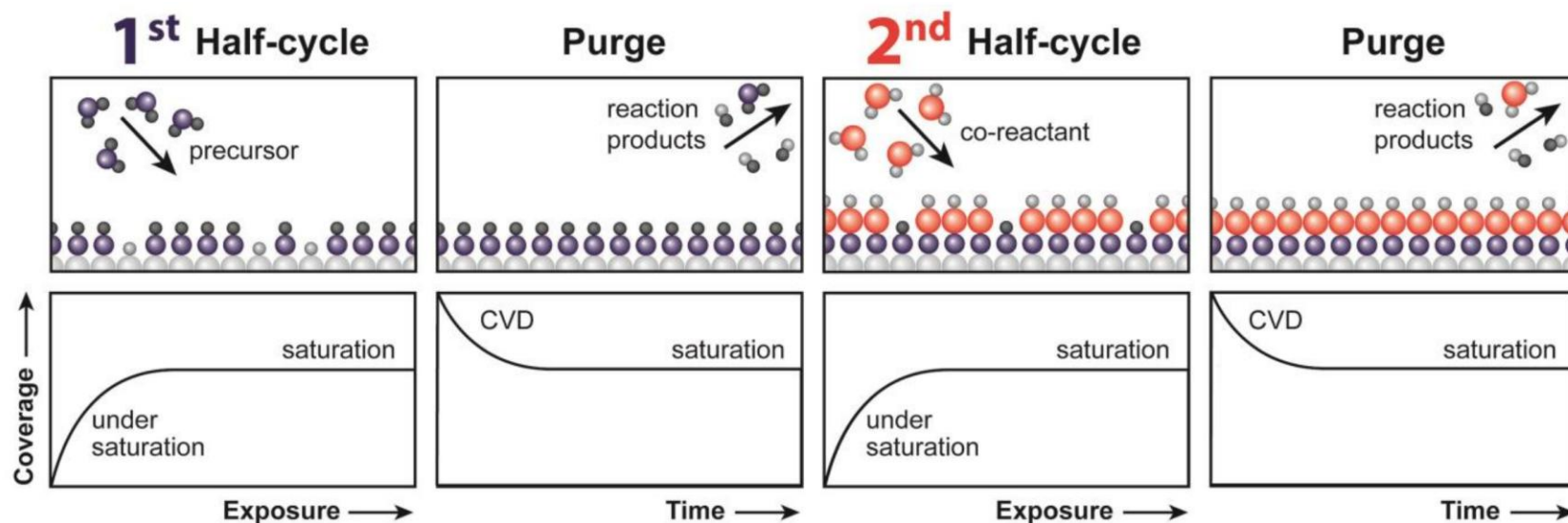


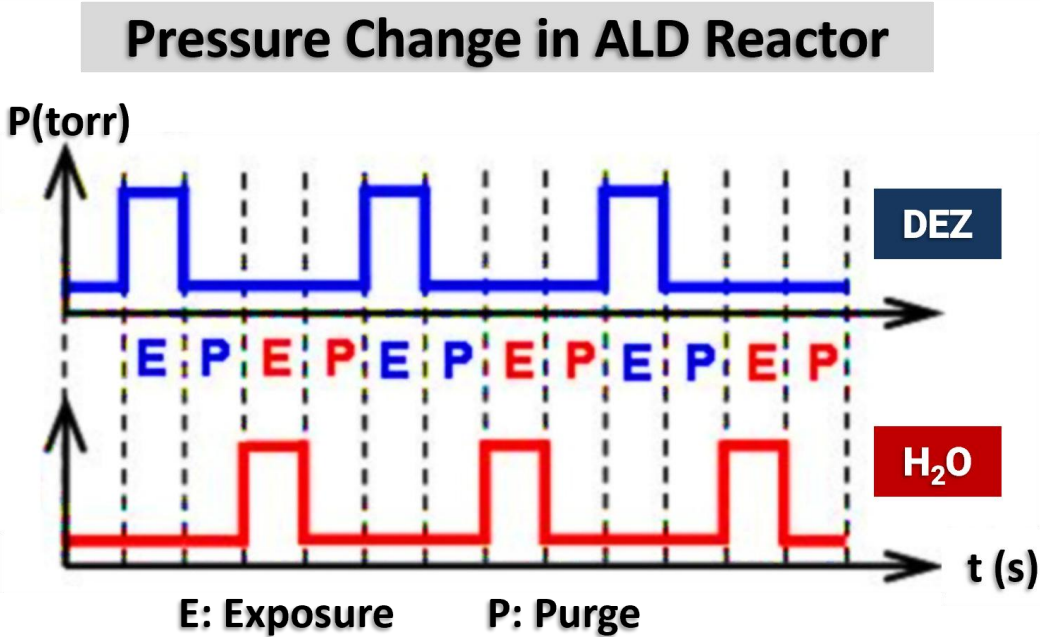
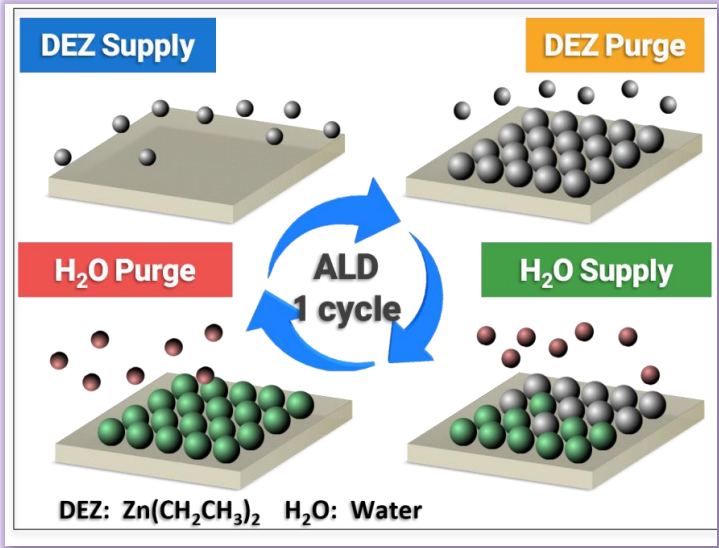
Fig. 3. A schematic representation of the various steps in an ALD cycle consisting of two half-reactions. The exposures in the first half-cycle (precursor) and second half-cycle (co-reactant) are self-limiting such that the process stops when all available surface sites are occupied. The two half-cycles are separated by purge steps. The half-cycles are repeated in an ABAB fashion to build the film up to the target thickness. The lower panels show the resulting coverage, or growth per cycle, as a function of exposure or time for that particular step. For sufficient exposure, saturated growth is obtained, while insufficient exposure results in incomplete saturation. For insufficient purging, a CVD component from mixing of the precursor and co-reactant is obtained.

H.C.M. Knoops, S.E. Potts, A.A. Bol, and W.M.M. Kessels, Ch. 27 - Atomic Layer Deposition (pp. 1101–1134) in Handbook of Crystal Growth, edited by T. Kuech (Elsevier, 2015).





Principle of ALD





Principle of ALD

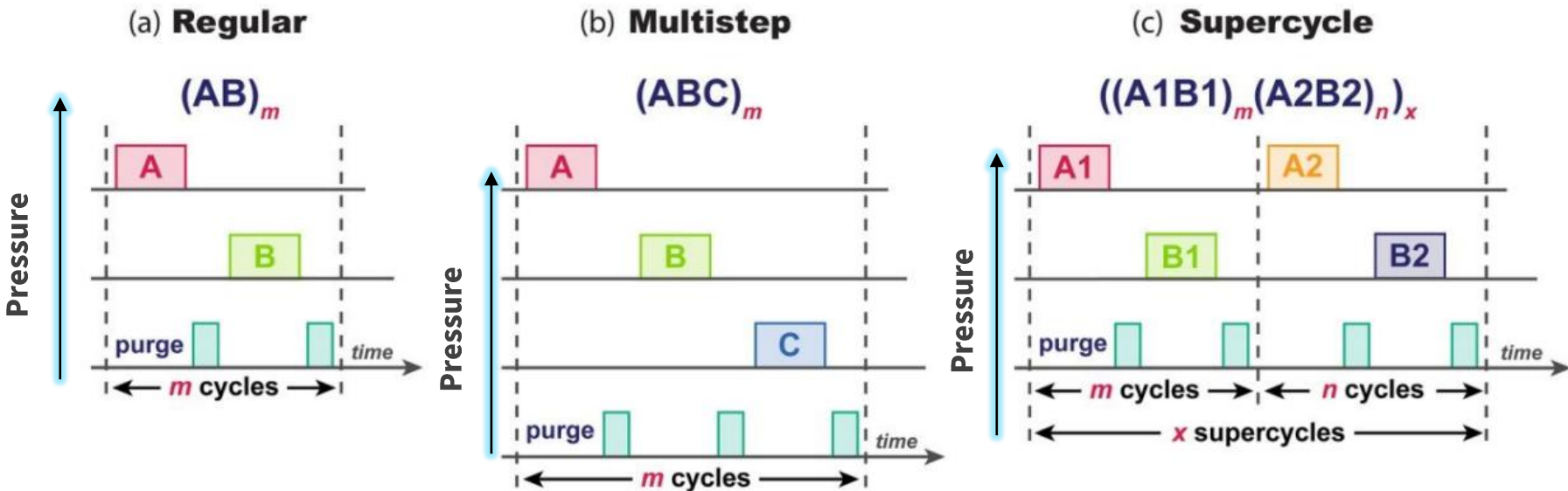


Fig. 6. A schematic representation of the various steps in (a) a regular ALD process, (b) a multistep process and (c) a supercycle. In a multistep process one or more additional steps are added in the ALD cycle to form for instance an ABC process. In a supercycle, the steps of two normal ALD processes are combined where m cycles of the first process are followed by n cycles of the second process. The variables m and n can be chosen so as to obtain the desired composition and structure of the film.



Principle of ALD

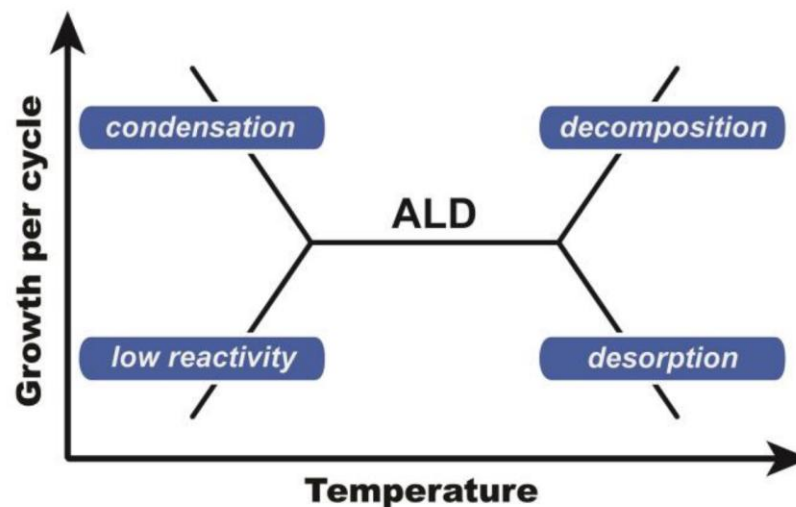
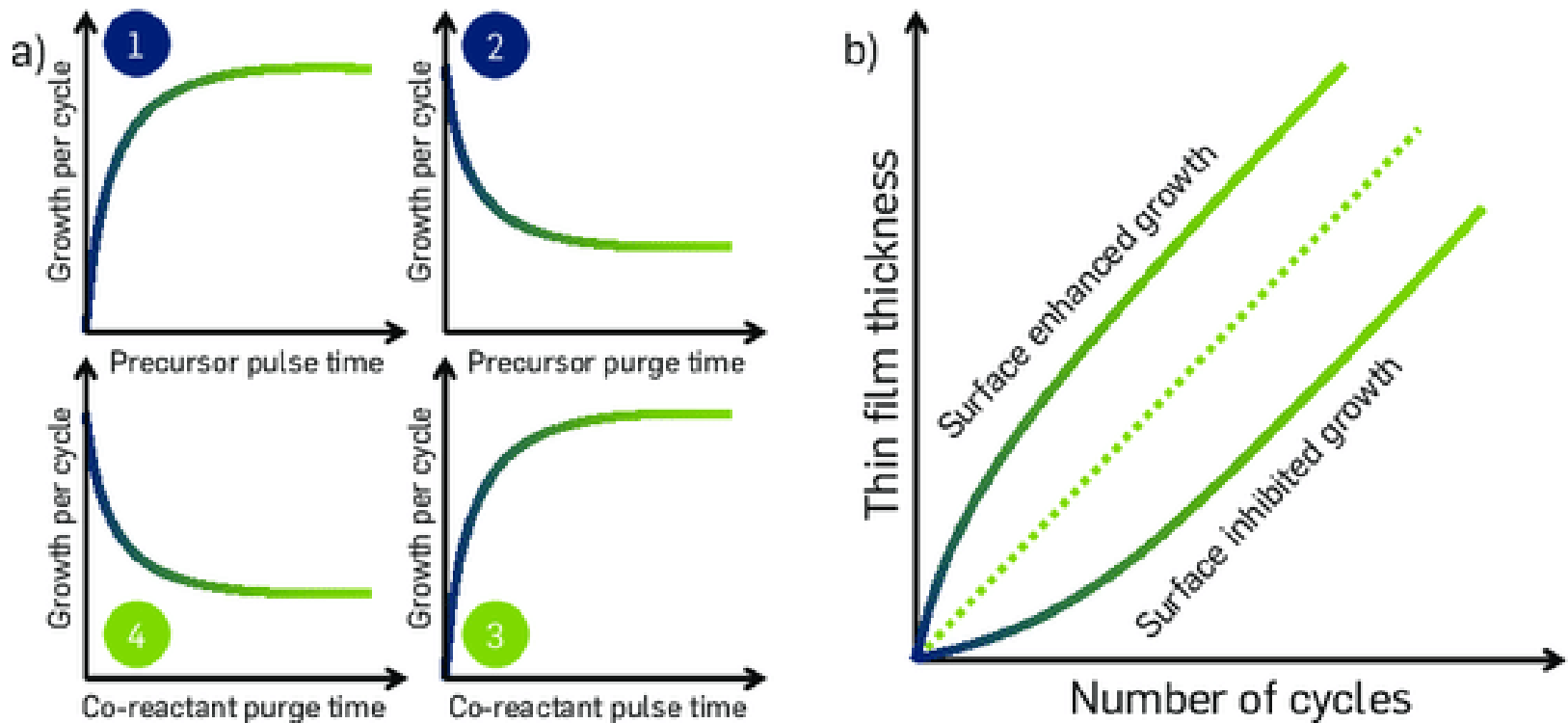


Fig. 4. The idealized temperature window of the ALD process, indicated by the growth per cycle as a function of temperature. In the case of ALD behavior, self-limiting growth is obtained and ideally a weak or no dependence on temperature is observed, while outside the window the ALD behavior is lost due to one of four effects. Condensation: precursors can condense on the surface preventing effective purging; low reactivity: low temperatures can prevent complete reactions from taking place; decomposition: when precursors decompose, a (parasitic) CVD component can be added to the growth; desorption: the film itself or the precursor can desorb, thereby reducing growth.





Principle of ALD

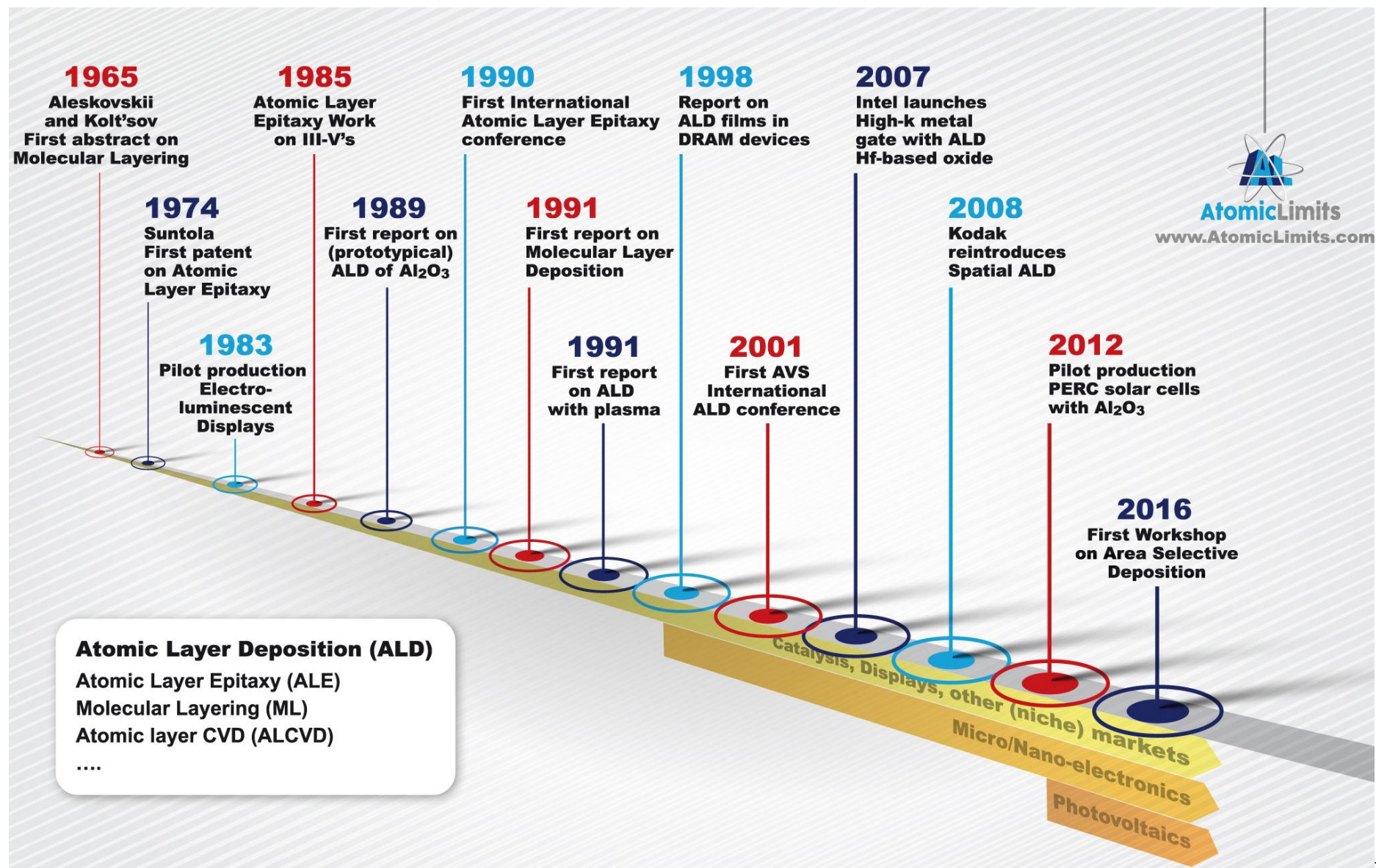


Typical ALD characteristics: a) Saturation: growth per cycle as a function of the (1) precursor pulse time, (2) precursor purge time, (3) co-reactant pulse time, and (4) co-reactant purge time; b) Linearity: thin film thickness as a function of the number of ALD cycles for a surface-enhanced and inhibited growth





Principle of ALD

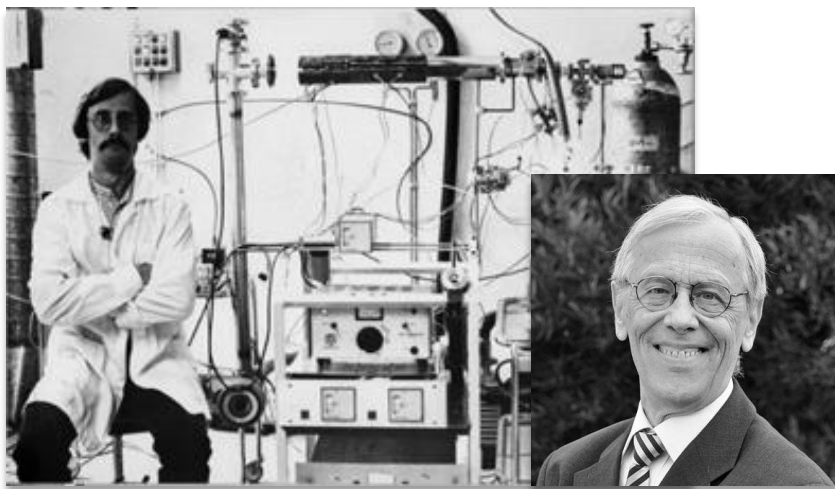




Inventor of ALD



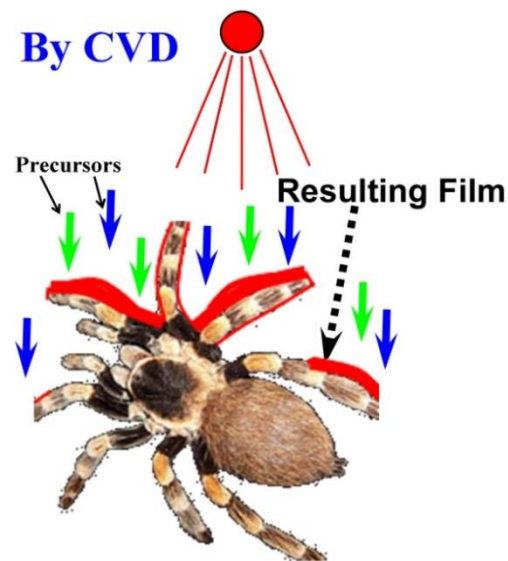
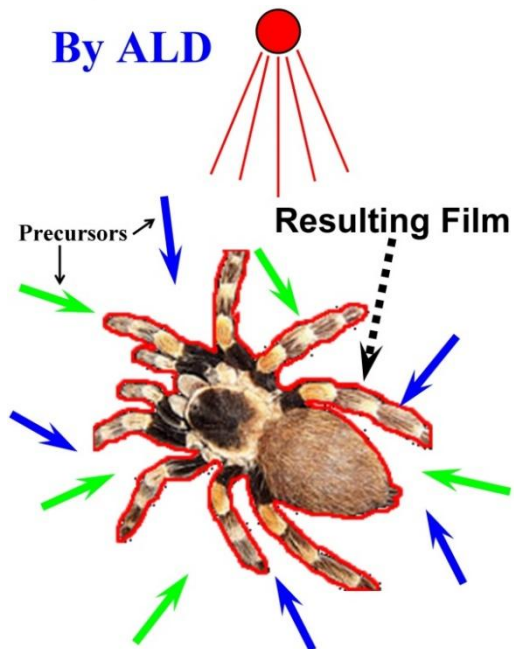
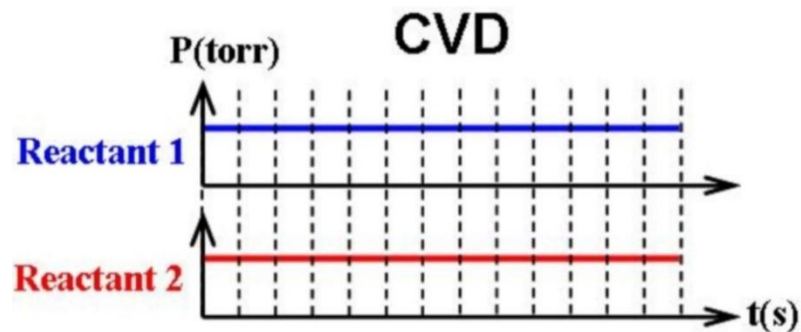
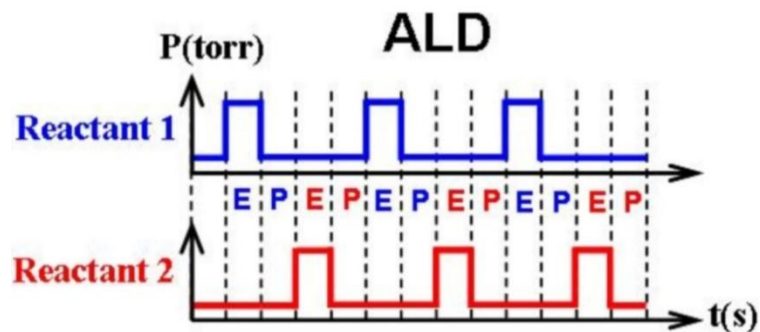
Tuomo Suntola(born 1943)



Sven Lindfors (1945-2017) in 1978 next to the flow-type ALD reactor in which the successful $\text{H}_2\text{S}/\text{ZnCl}_2$ process was demonstrated.

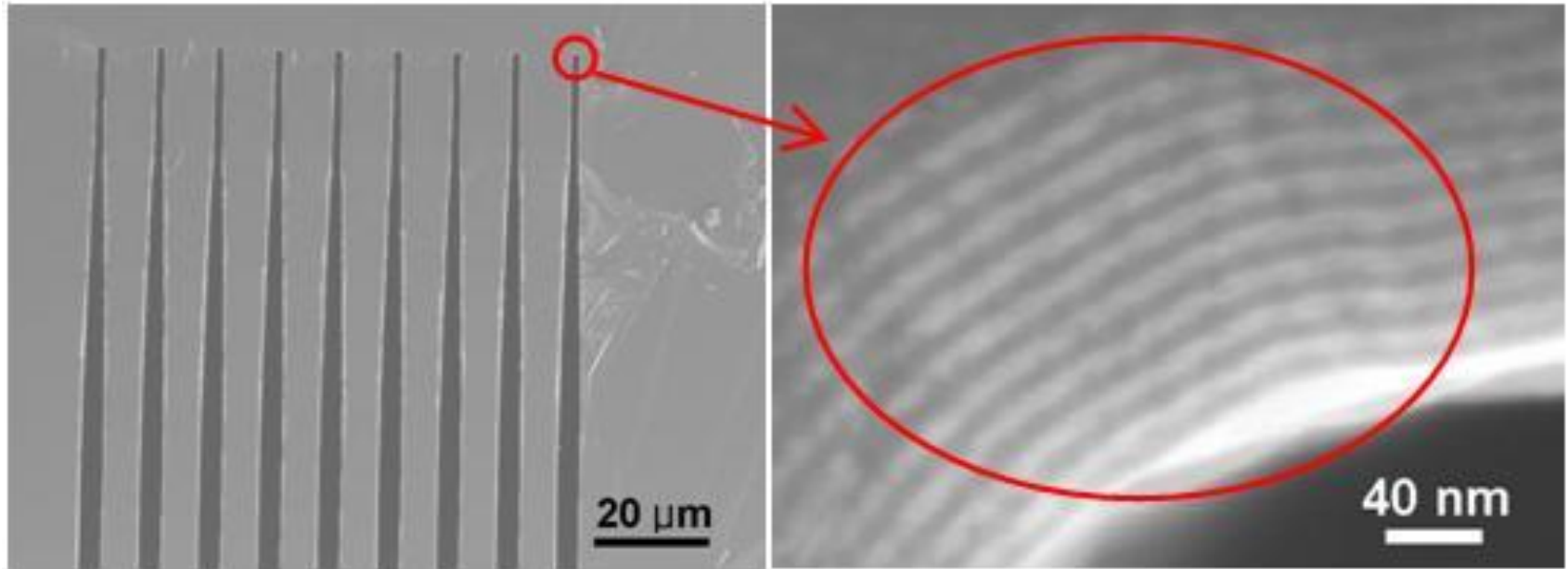


Deposition feature of ALD and CVD





Step coverage

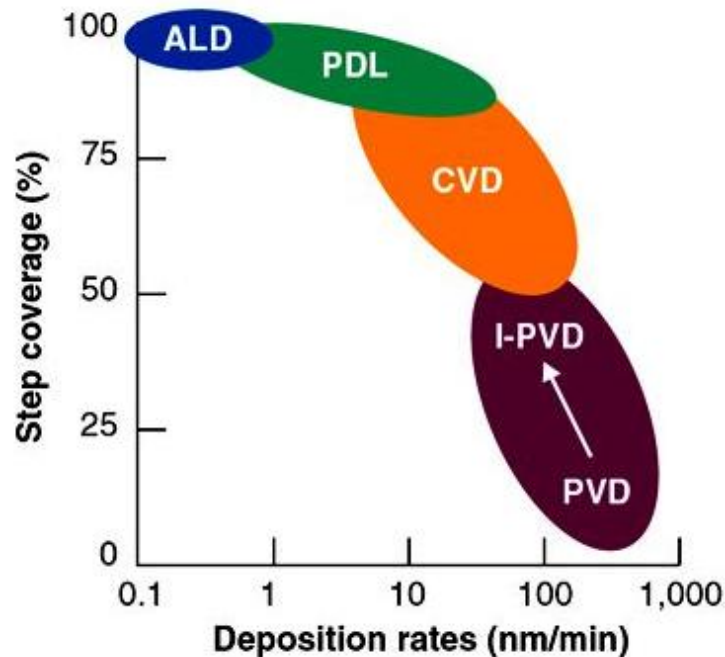


ALD sample by Riikka Puurunen

ALD vs other tools



ALD: slowest but best step coverage



PLD: Pulsed Deposition Layer; I-PVD: Ionized physical vapor deposition

Advantage

- Stoichiometric films with large area uniformity and 3D conformality.
- Precise thickness control and low-temperature deposition are possible.
- Gentle deposition process for the sensitive substrate.

Disadvantage

- The deposition rate is slower than CVD
- Number of different materials that can be deposited is fair compared to MBE



Overview of materials grown by ALD by Feb 2005

Actinoids**

-18-



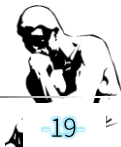
Atomic layer deposition

45th anniversary 1974 - 2019



<https://www.atomiclimits.com/2019/11/28/atomic-layer-deposition-turns-45-the-history-of-ald-in-a-timeline-and-an-animated-version-of-the-ald-periodic-table/>

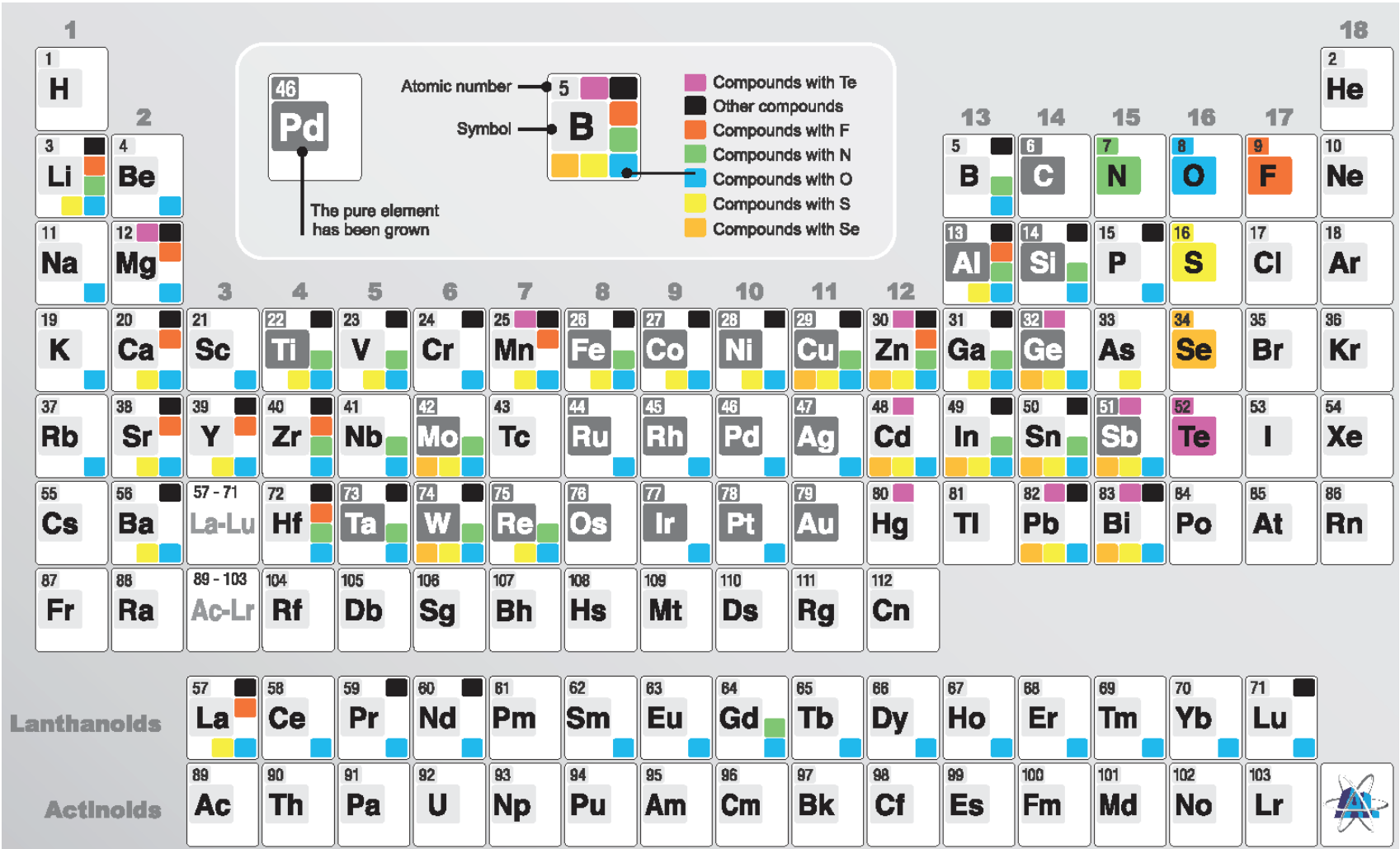
<https://www.atomiclimits.com/alddatabase/>





ALD Materials

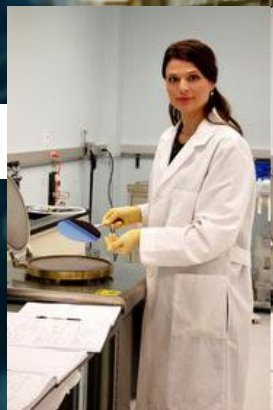
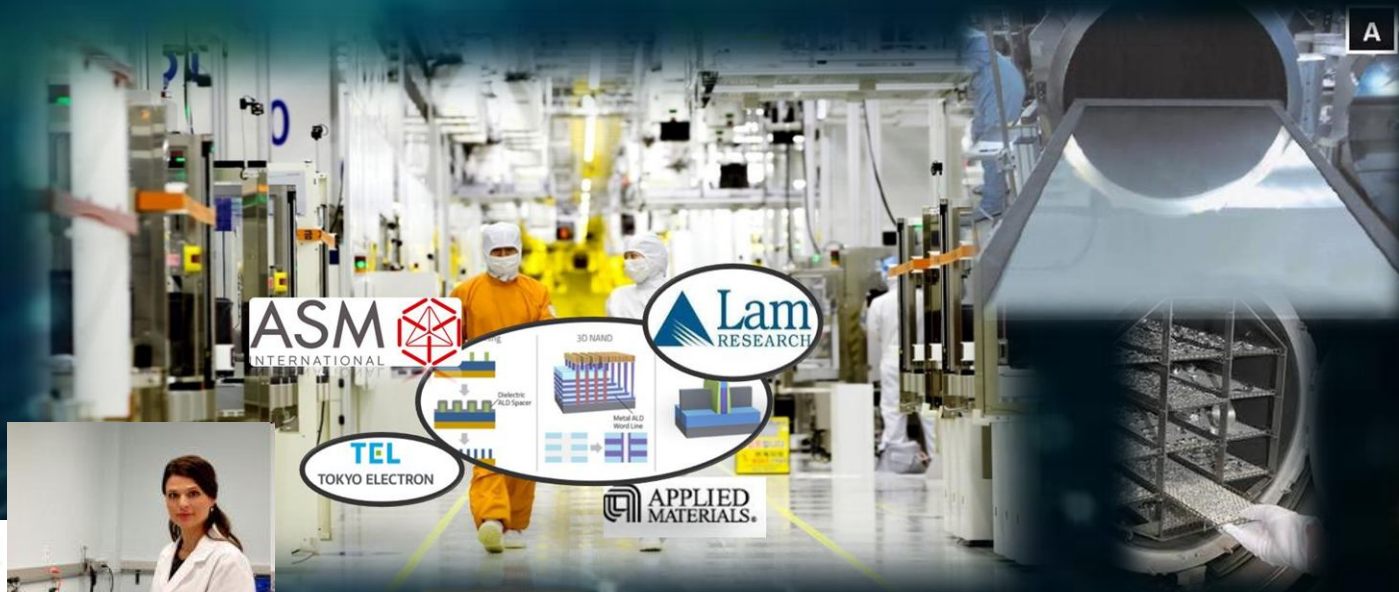
Overview of materials grown by ALD by Feb 2019



<https://www.atomiclimits.com/alddatabase/>



ALD Manufacturers



And many other companies...

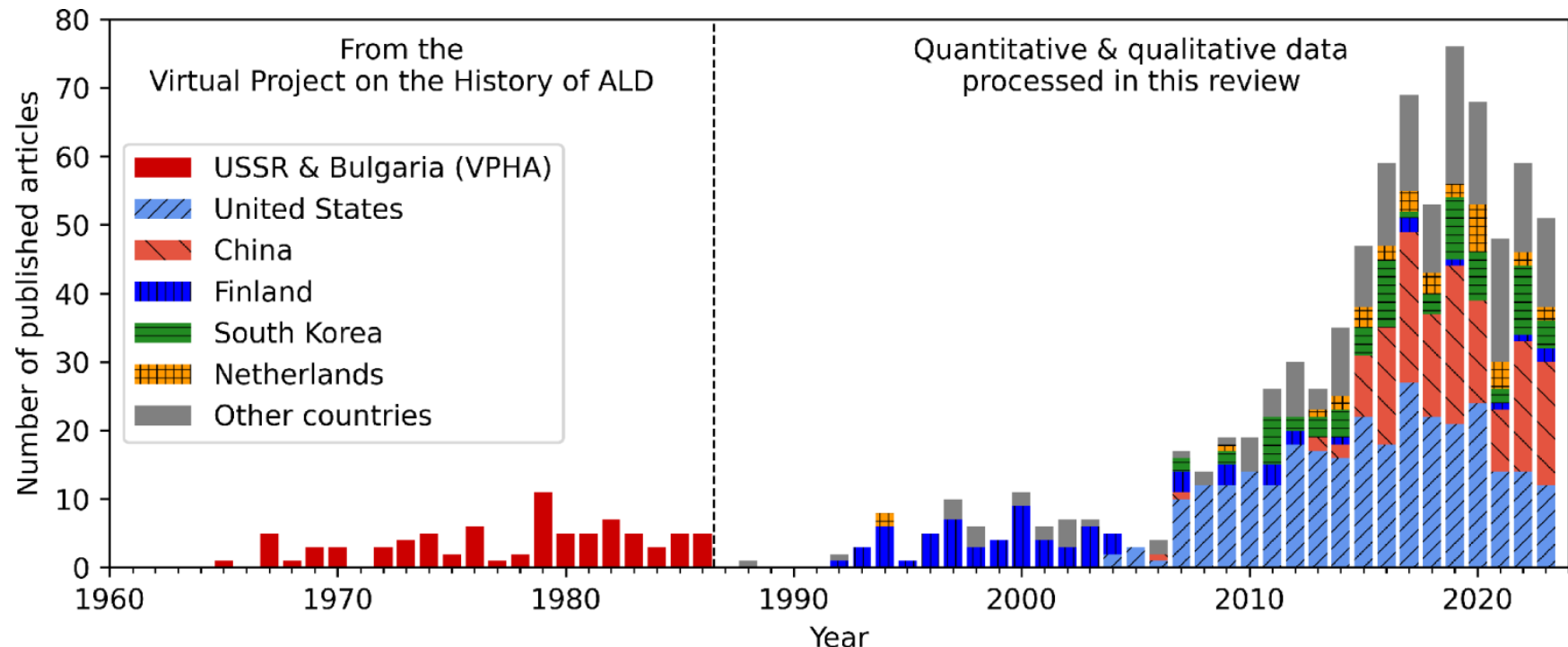


Figure 1. Number of published articles per year. The quantitative analysis in the present review starts from the earliest Japanese article from the year 1988,¹ and is shown per country for the top five contributors, in order of number of contribution. Earlier contributions were identified in the Virtual Project on the History of ALD (VPHA), and came from the USSR and Bulgaria (investigation up to 1986, dashed line).





ALD Applications?

Before 2010

Semiconductors & nanoelectronics

Gate dielectrics
Gate electrodes
Ferroelectric
Non-volatile Memory
Metal Interconnects
Diffusion barriers
DRAM
Multilayer-capacitors
Read heads

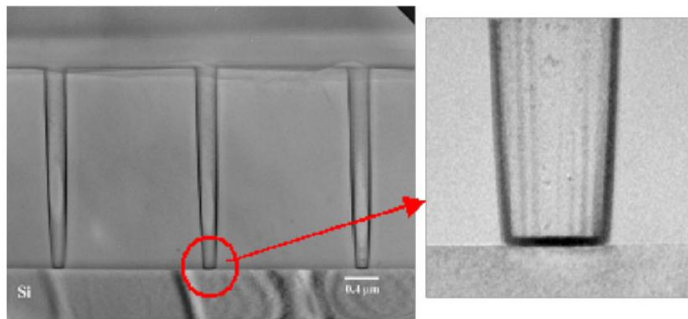


Figure 11. Scanning electron microscope image of trench structure showing excellent coverage by ALD HfO_2 . Thanks to McIntyre.

Rep. Prog. Phys. **69**, 327–396 (2006)

Optical

Antireflection
Optical filters
OLED layers
Photonic crystals
Transparent conductors
Electroluminescence
Solar cells
Lasers
Integrated optics

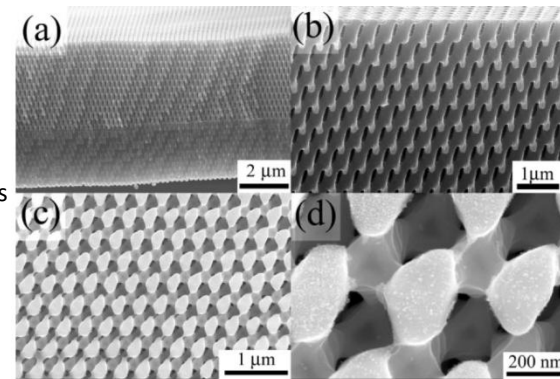
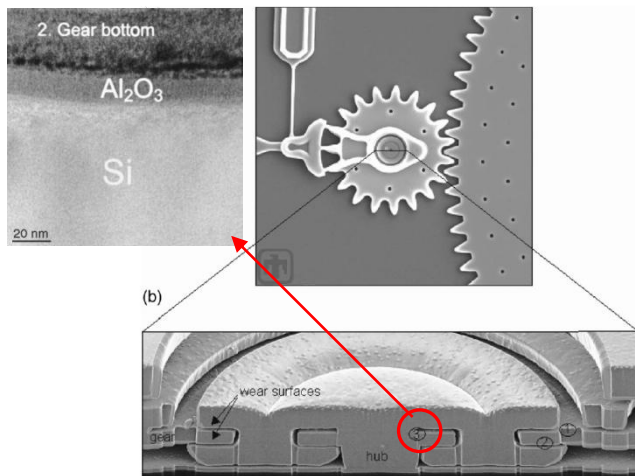


Figure 2. Inverted TiO_2 photonic crystal formed after template removal by O_2 plasma etch. a,b) Cross sections revealed by cleavage along a (111) plane. c,d) Top (111) surface of film.

Adv. Mater. **18**, 1561–1565, 2006

MEMS

Wear resistance



Appl. Phys. Lett. **82**, 2883–2885, 2003

Nanostructures

Nanoparticles
Nanotubes

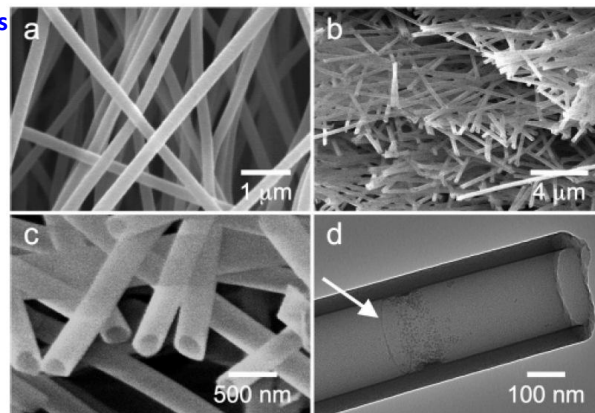


Figure 2. (a) FESEM image of electrospun PVA fibers. (b, c) FESEM images of Al_2O_3 microtube replicas prepared by ALD wherein the Al_2O_3 coating was deposited for 475 cycles at 45°C . (d) TEM image of a corresponding Al_2O_3 microtube. The arrow in (d) identifies a peripheral circle formed inside the microtube near an apparent neck in the fiber template.

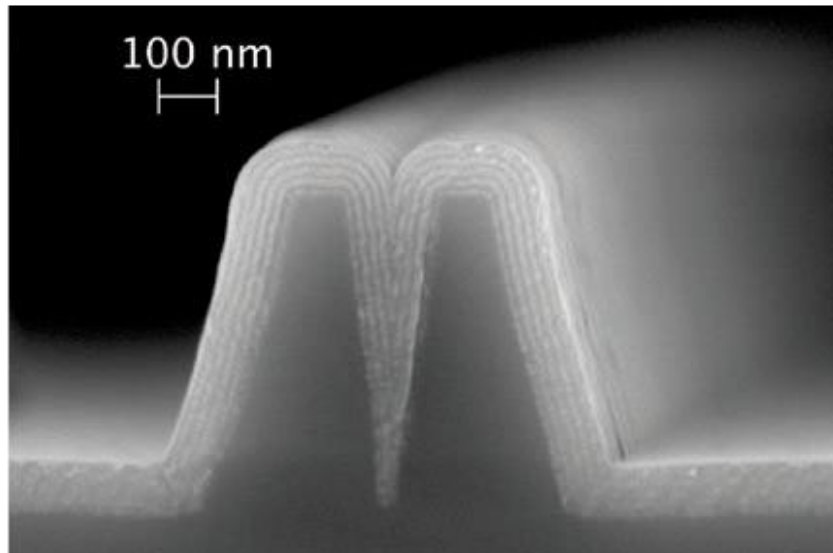
Nano Lett. **7**, 719–722, 2007



ALD Applications ?



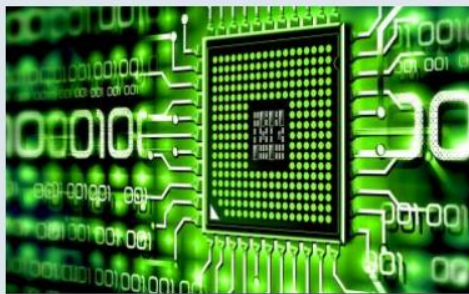
In 2012



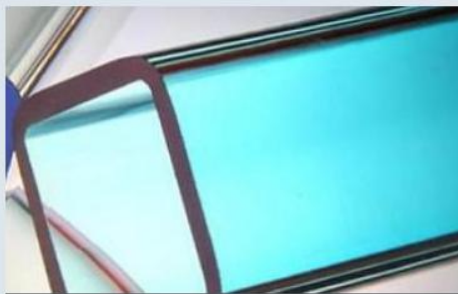


ALD Applications?

In 2019



More than Moore (MtM)



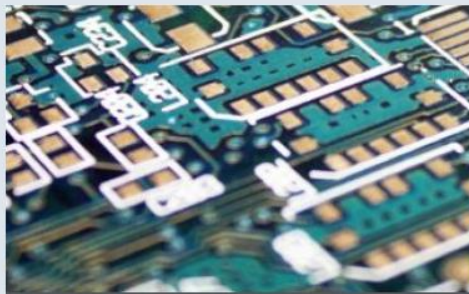
Optical coatings



Moisture barriers



Anti-tarnish coatings



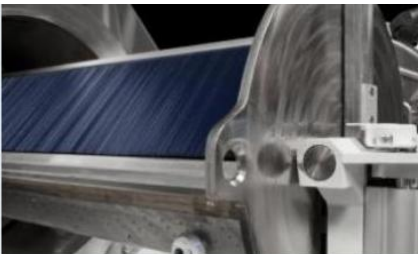
Anti-corrosion coatings



Conductive coatings



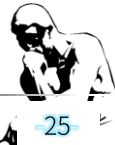
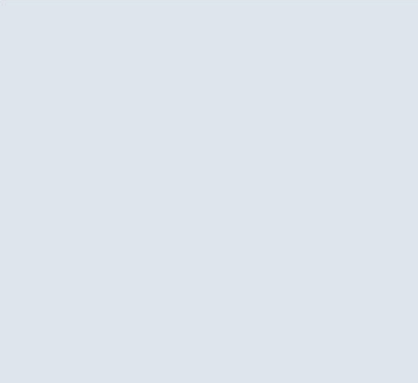
Insulators



Passivation



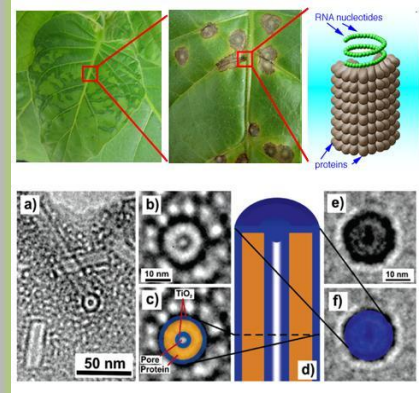
Decorative coatings



Some Unique Applications of ALD-Biotemplate

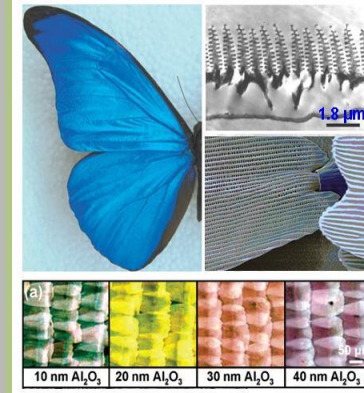


ALD on Tobacco Mosaic Virus



Nano Lett. **6**, 2325-2331 (2006)

ALD on Butterfly Morpho



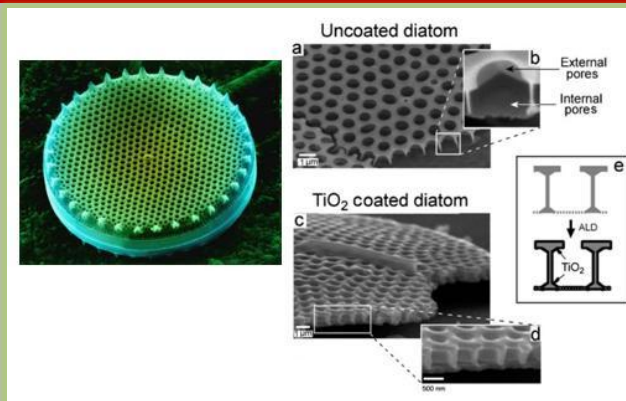
Nano Lett. **6**, 2325-2331 (2006)

ALD on Fly's eye



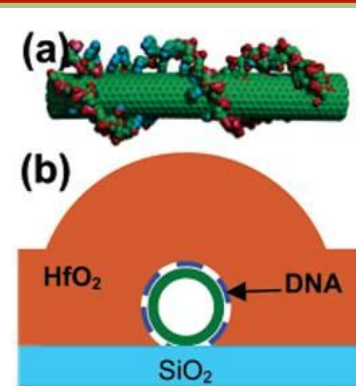
J. Am. Chem. Soc. **128**, 3518-3519 (2006)

ALD on Diatom



J. AM. CHEM. SOC. **128**, 3518-3519 (2006)

ALD on DNA

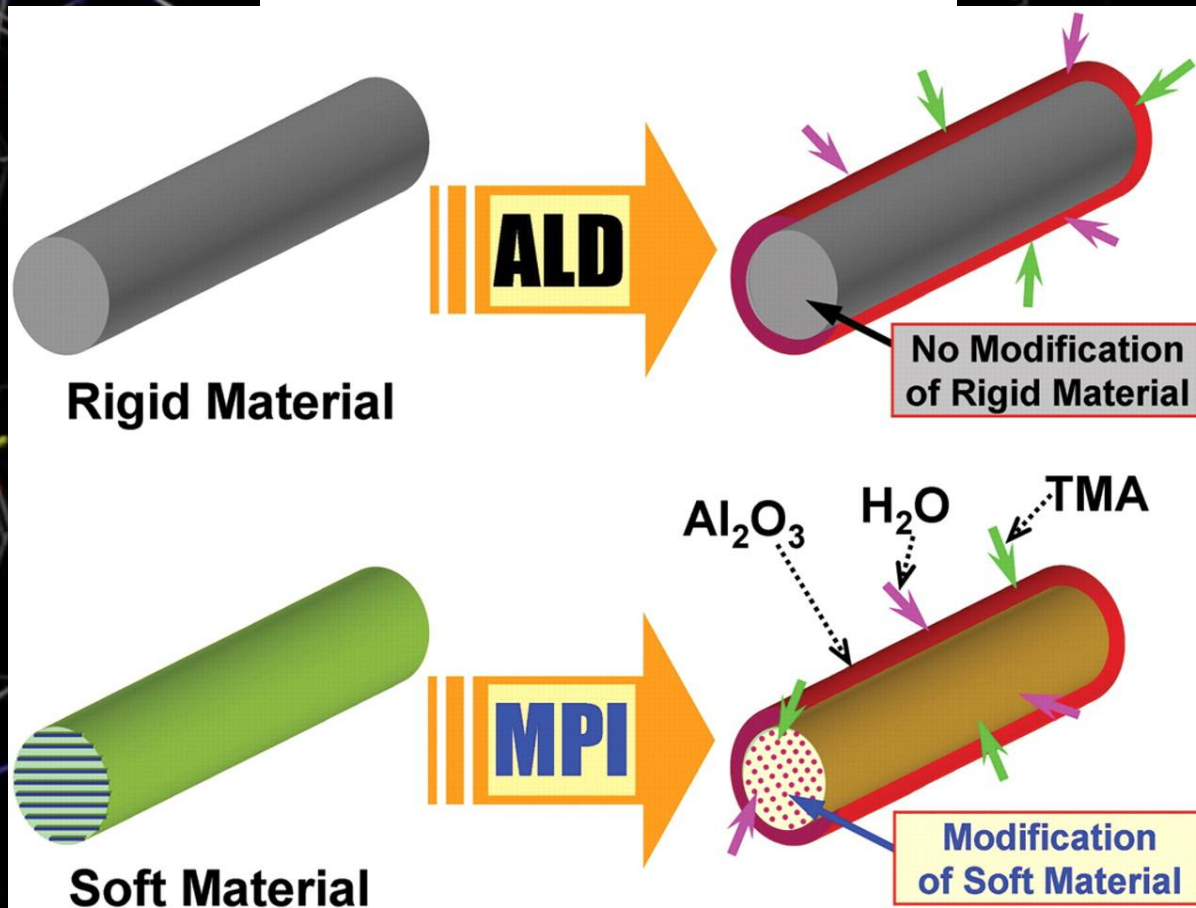


J. Am. Chem. Soc. **128**, 3518-3519 (2006)

*VPI: Vapor Phase Infiltration

Greatly Increased Toughness of Infiltrated Spider Silk

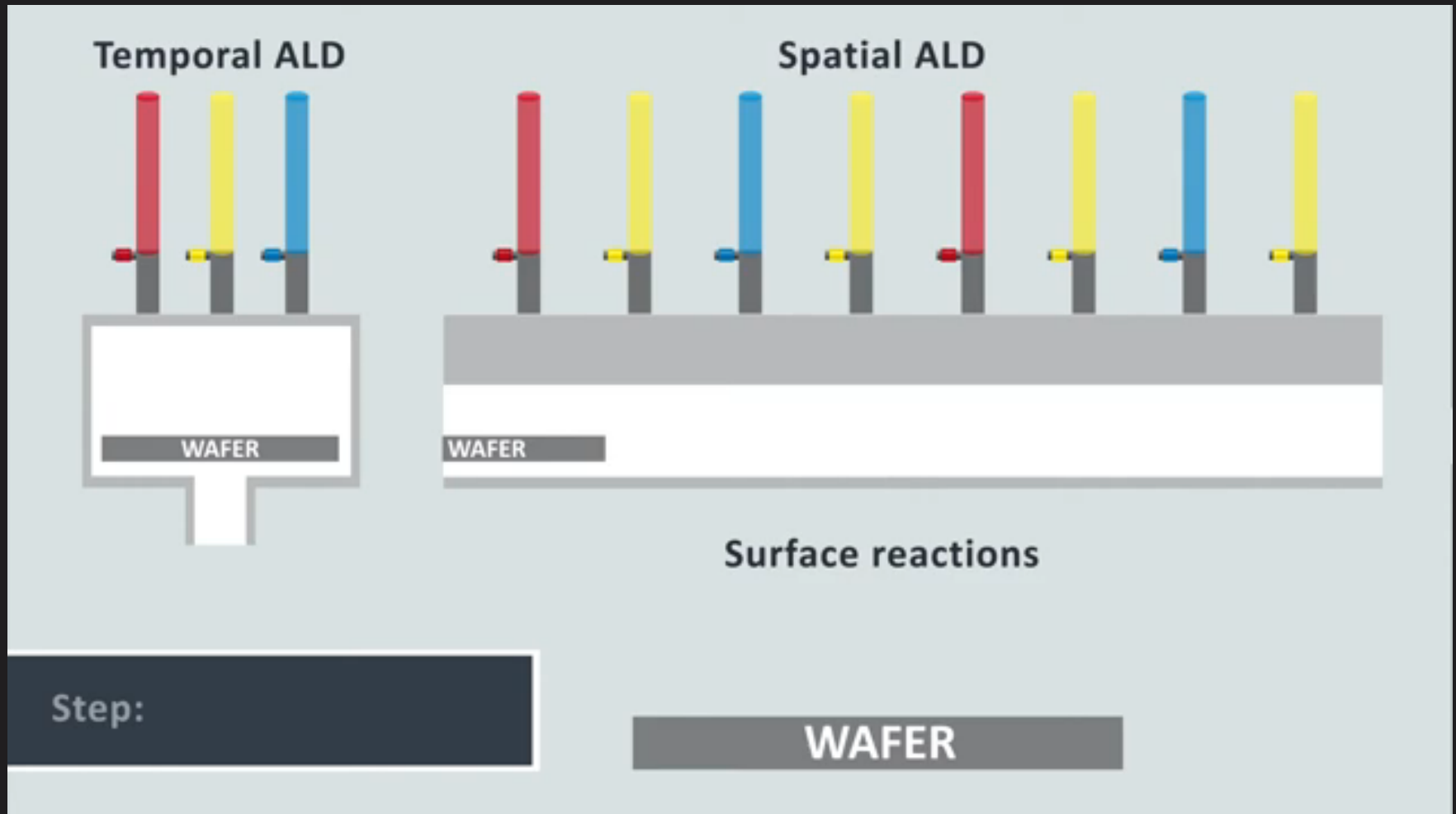
Seung-Mo Lee,^{1*} Eckhard Pippel,¹ Ulrich Gösele,¹ Christian Dresbach,² Yong Qin,¹ C. Vinod Chandran,³ Thomas Bräuniger,³ Gerd Hause,⁴ Mato Knez^{1*}

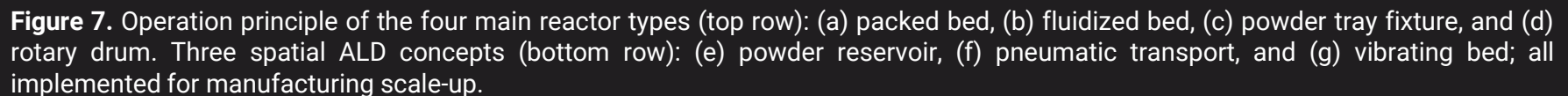


시간분할 ALD vs 공간분할 ALD



시간분할 ALD vs 공간분할 ALD





30

시간분할 ??? 공간분할 ???

시간분할 ALD



공간 분할 ALD



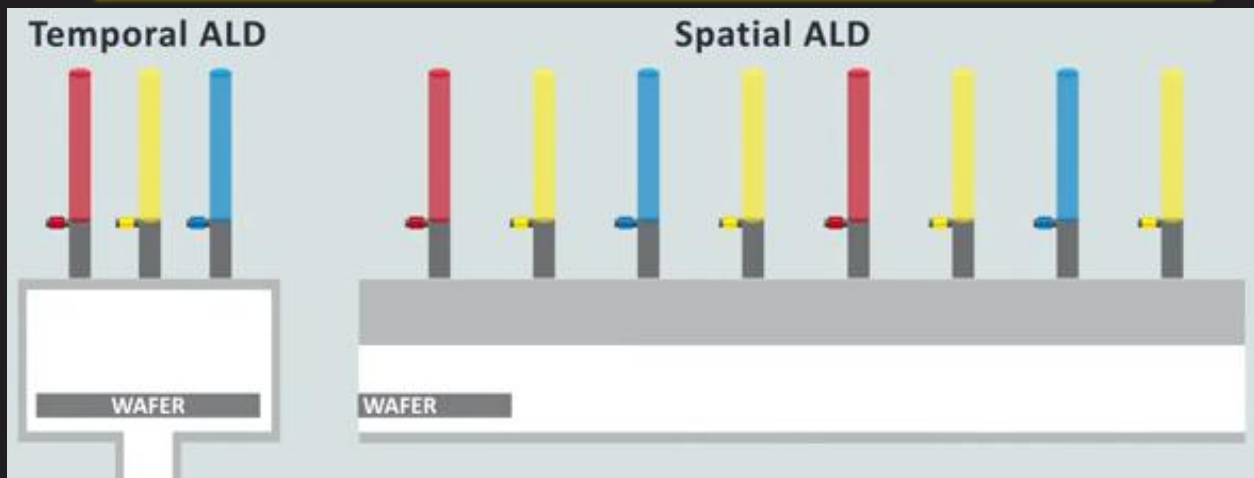
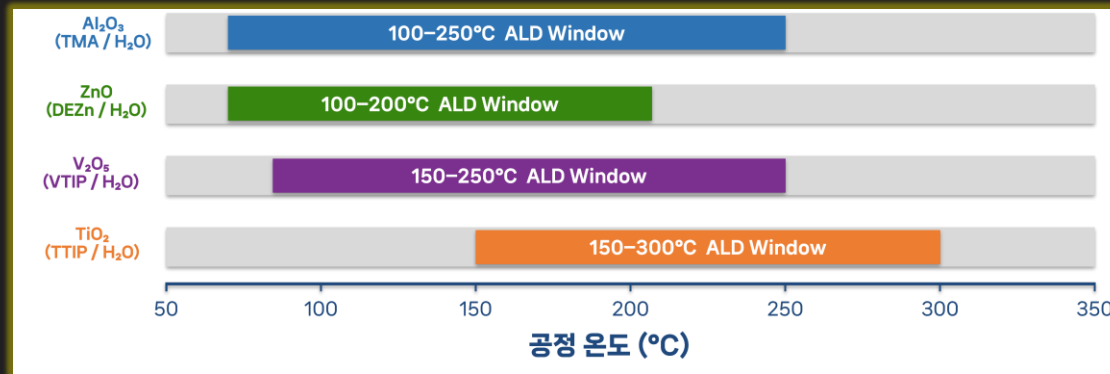
I 시간분할 ALD vs 공간분할 ALD I

구분	시간분할 ALD (Temporal ALD) @KIMM	공간분할 ALD (Spatial ALD) @대성기계공업
전구체 분리 원리	1번 전구체가 챔버로 유입되어 표면과 반응한 뒤, 2번 전구체를 흘리기 전에 챔버에서 퍼지된다.	1번과 2번 전구체가 동시에 챔버로 흘러가되 공간적으로 분리되어, 두 흐름 사이에 혼합을 막는 영역이 존재한다.
기판/전구체의 상대 운동	기판 고정, 가스가 시간 순서대로 교체	기판이 가스 분배판에 대해 상대적으로 이동하거나 그 반대로 이동한다.
퍼지(purge) 단계	챔버를 퍼지하는 데 상당한 시간이 필요할 수 있으며, 실제로는 추가 반응물 유입 전 잔류 반응물을 완전히 제거하기 어려운 경우도 있다.	공간적 분리로 인해 여분의 반응물을 퍼지할 필요가 없고 교차오염이 제한되므로, 퍼지 단계를 없앴으로써 스루풋을 높일 수 있다.
압력 조건	주로 진공	가장 널리 쓰이는 형태는 진공이 필요 없는 대기압 공간분할 ALD(일명 Atmospheric ALD)
핵심 트레이드오프	긴 정체시간 → 균일도/컨포멀리티 우수, 낮은 스루풋	전구체가 잘 분리되어 기판(또는 이전 증착층)하고만 반응하면 균일하고 컨포멀한 ALD 영역에서 성장이 일어나지만, 전구체 혼합이 심해지면 빠르지만 대체로 덜 균일한 CVD 영역으로 넘어간다.

목 차

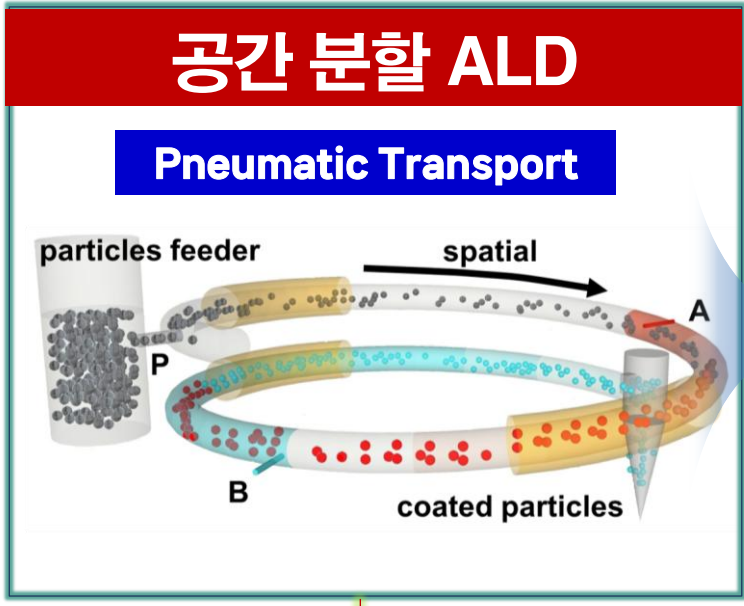
화제 2.

1-3차년도 연구 계획

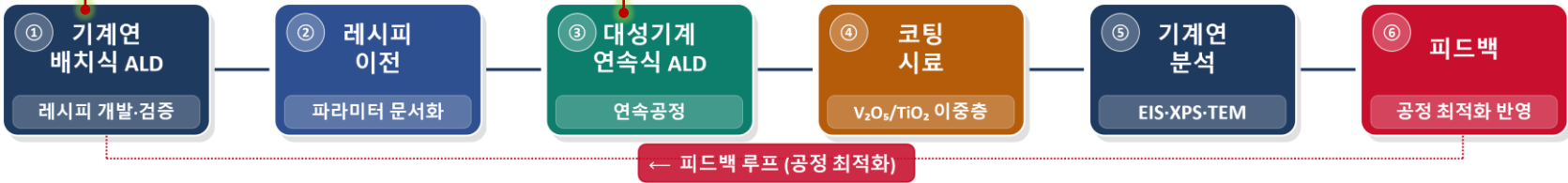




진공 필요
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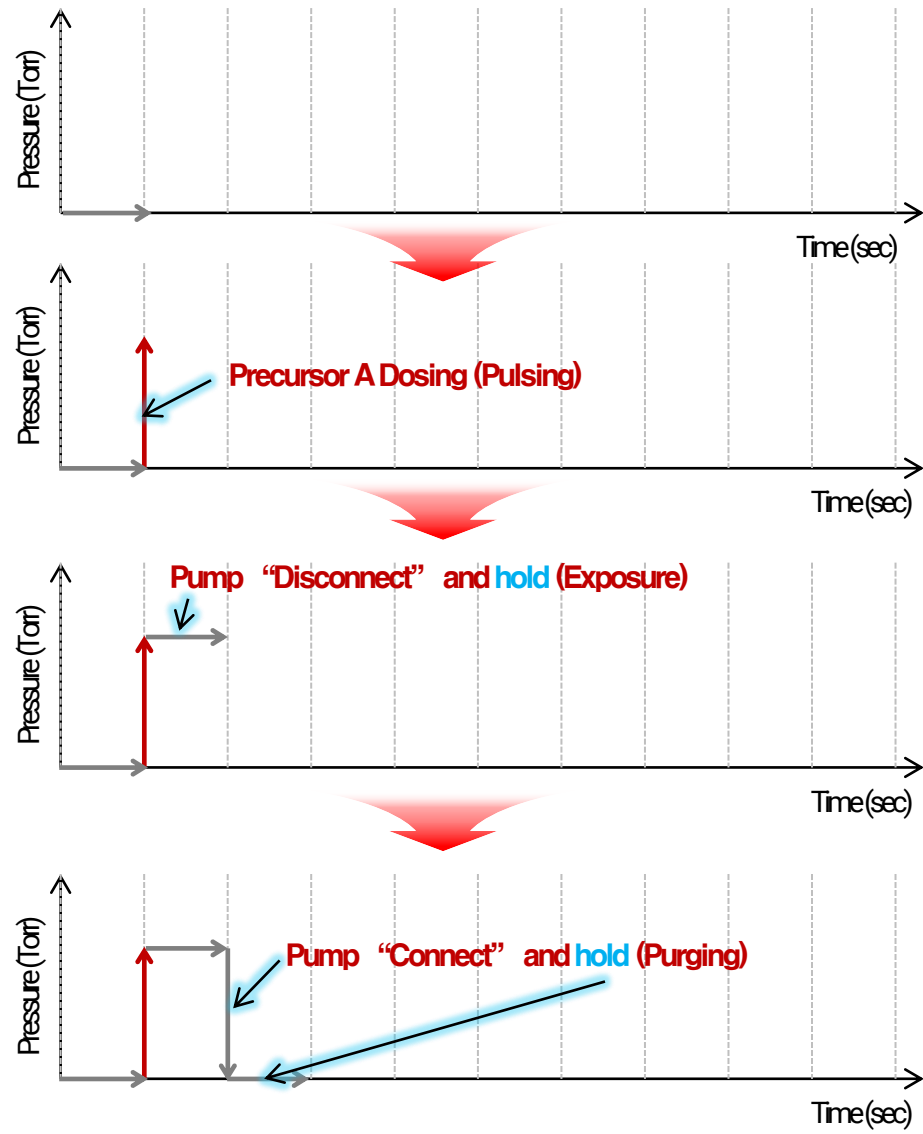
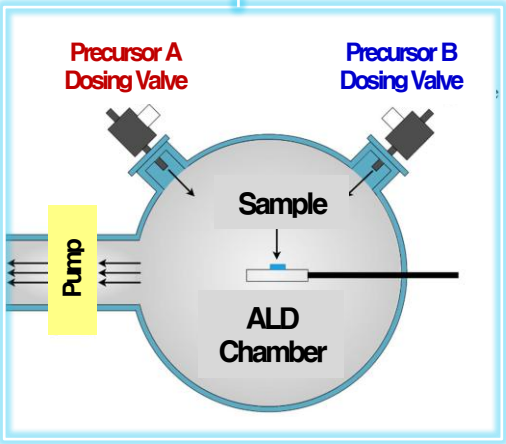


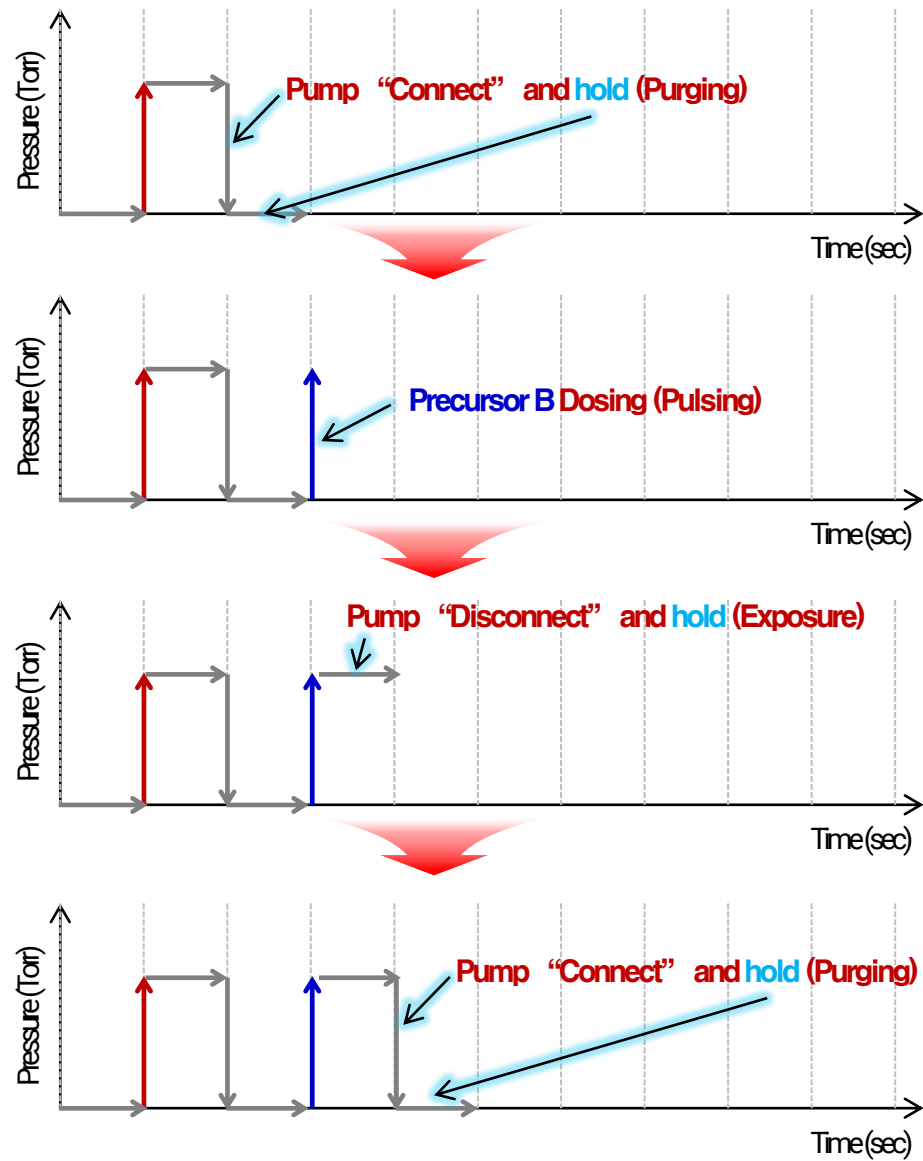
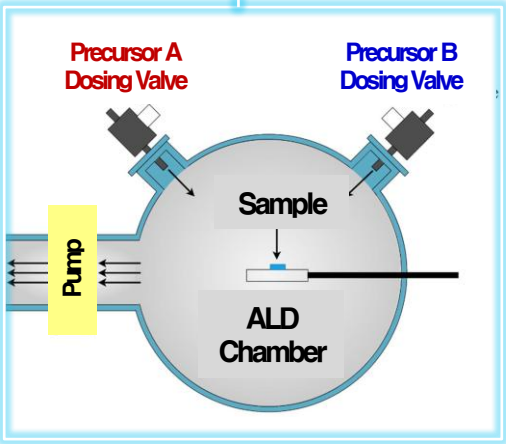
진공 불필요
(~760 Torr)

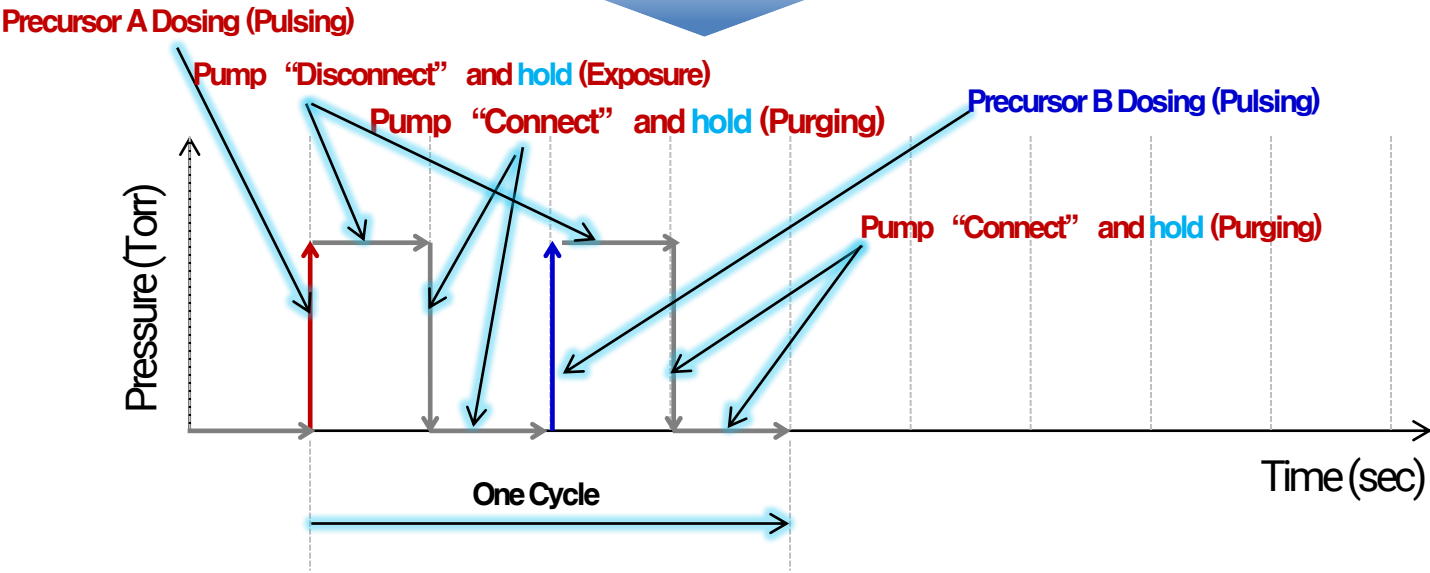
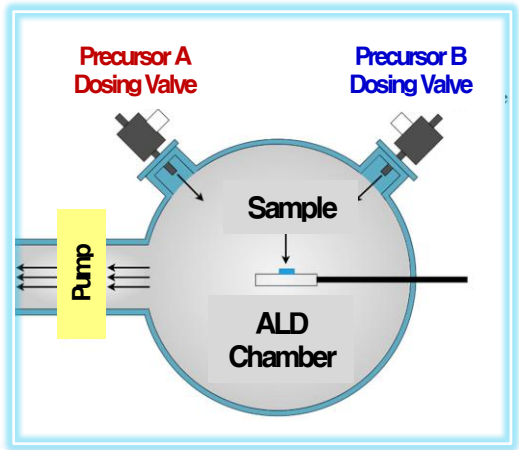




시간 분할 ALD

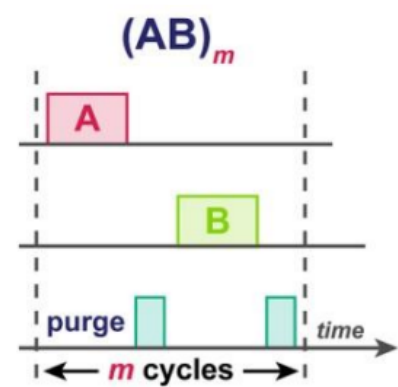




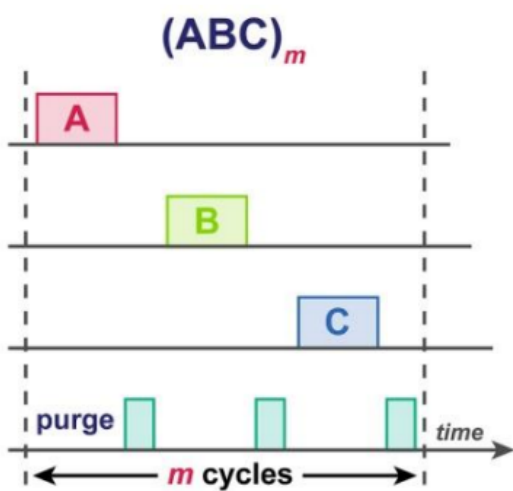




Regular



Multistep



Supercycle

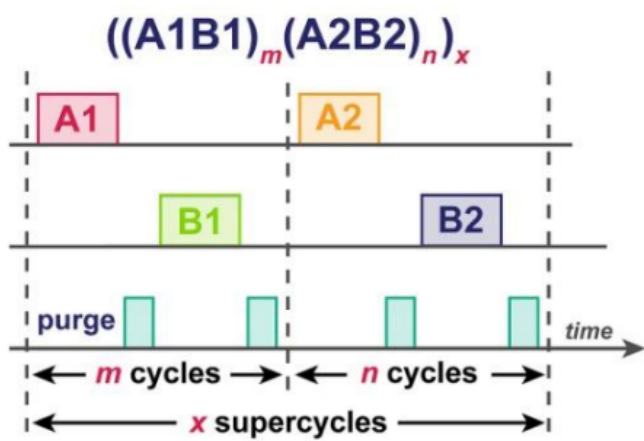
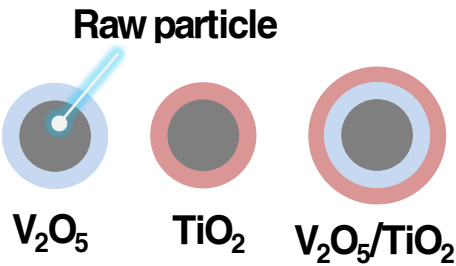
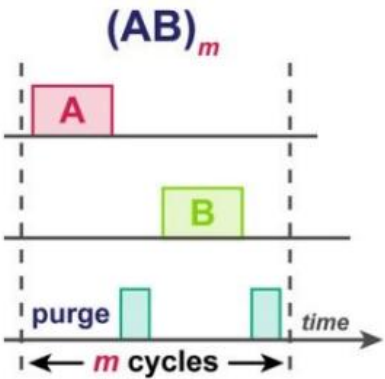


Fig. 6. A schematic representation of the various steps in (a) a regular ALD process, (b) a multistep process and (c) a supercycle. In a multistep process one or more additional steps are added in the ALD cycle to form for instance an ABC process. In a supercycle, the steps of two normal ALD processes are combined where m cycles of the first process are followed by n cycles of the second process. The variables m and n can be chosen so as to obtain the desired composition and structure of the film.

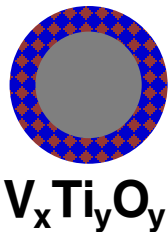
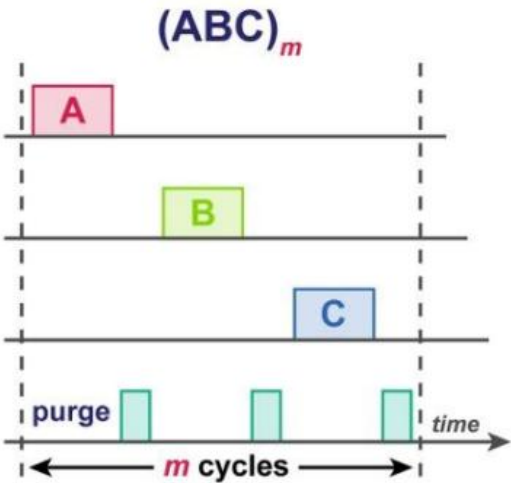




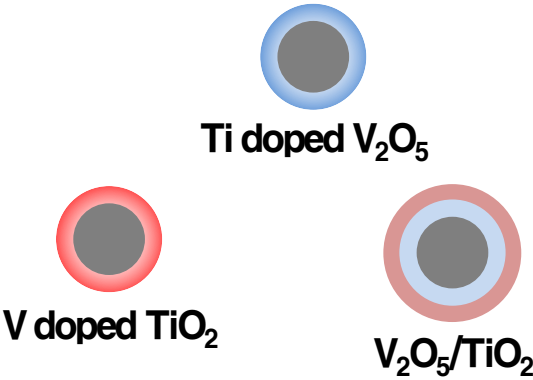
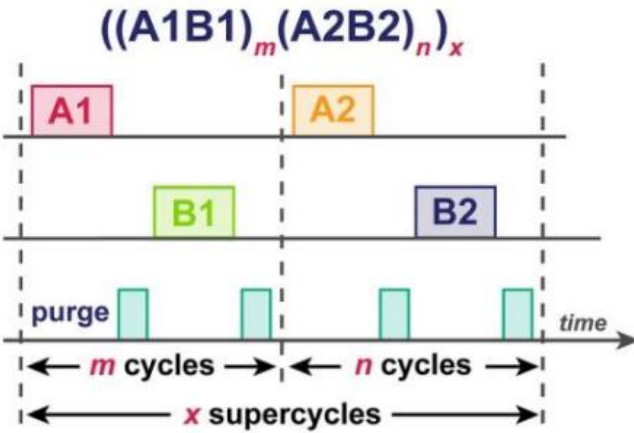
Regular

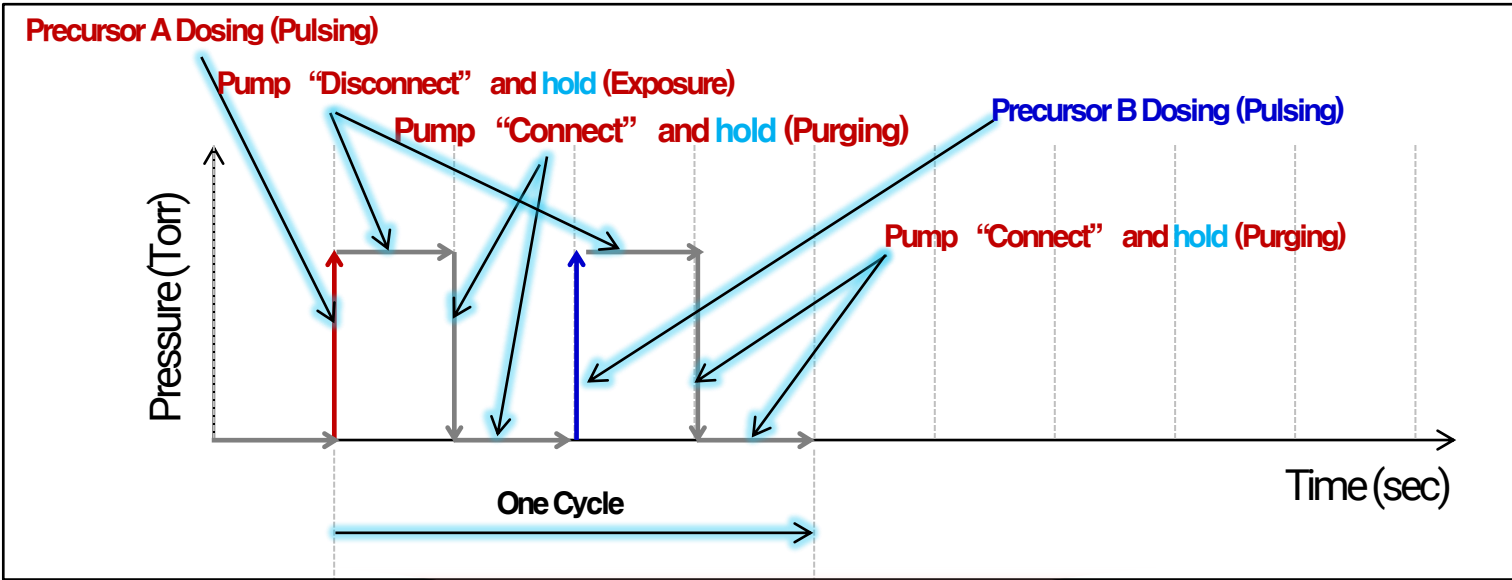


Multistep



Supercycle





- Key 01: 어떤 산화물로 처리를 할까?
- Key 02: 어떤 전구체를 써야 할까?
- Key 03: 각 전구체의 Pulse time, Exposure time, Purge time?
- Key 04: 각 전구체의 가열 온도 ? 챔버의 온도는?
- Key 05: 증착이 이루어 질까? 결정화도는? 코팅의 안정성은?
- Key 06: ALD 처리가 된 양극재의 안정성이 확보될 수 있을까? 나트륨전지가 잘 구동될까?
- Key 07: 시간분할 ALD로 확보된 레서피를 공간분할 ALD에 어떻게 적용해야 할까?





Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

Surface degradation mechanisms in Na-layered oxides



HF 공격 [1]

전해질 분해 HF가 양극 표면 용해 → AlF_3 보호층으로 차단 필요



비가역 상전이 [2]

$\text{P2} \rightarrow \text{O2}$, $\text{O3} \rightarrow \text{P3}$ 상전이 시 부피 변화 → 입자 크랙·용량 감소



전이금속 용출 [1,2]

Mn, Ni 용출 → 음극 오염 및 계면 임피던스 증가



Na_2CO_3 표면 잔류 [2]

공기 노출 시 Na_2CO_3 형성 → 계면 저항 증가

1) Ji, G. et al., Improvement of the Cathode Electrolyte Interphase on $\text{P2-Na}_{2/3}\text{Ni}_{1/3}\text{Mn}_{2/3}\text{O}_2$ by Atomic Layer Deposition (ACS Appl. Mater. Interfaces, 2017)
2) Sun, L. et al. Recent progress of interface modification of layered oxide cathode material for sodium-ion batteries (Electron (Wiley / HIT), 2024)





Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

O3형 [2,3]

NaTMO₂ (x ≈ 1)

배위: Na: 팔면체(Octahedral)
적층: ABCABC (3겹 반복)

▲ 장점
 초기 용량 높음 (~150 mAh/g)
 높은 Na 함량
 합성 용이

▼ 단점
 P'2·O1 상전이 심함
 사이클 안정성 낮음
 공기 민감성 높음

NaNi_{0.5}Mn_{0.5}O₂
 NaNi_{0.6}Co_{0.2}Mn_{0.2}O₂

P2형 [1,2]

Na_{2/3}TMO₂ (x ≈ 0.67)

배위: Na: 삼각프리즘(Prismatic)
적층: ABBA (2겹 반복)

▲ 장점
 사이클 안정성 우수
 고전압 안정
 Na⁺ 확산 빠름

▼ 단점
 초기 용량 제한 (~130 mAh/g)
 Na 함량 낮음
 P2→O2 고전압 전이

Na_{2/3}[Ni_{1/3}Mn_{2/3}]O₂
 Na_{2/3}Ni_{0.3}Co_{0.1}Mn_{0.6}O₂

P3형 [3]

Na~0.5TMO₂ (x ≈ 0.5)

배위: Na: 삼각프리즘(Prismatic)
적층: ABBCA (3겹)

▲ 장점
 합성 유연성 높음
 다양한 조성 가능
 중간 수준 용량

▼ 단점
 O3·P2보다 연구 부족
 구조 안정성 중간
 상전이 거동 복잡

Na_{0.5}TMO₂ 계열
 (Ti, Mn, Cr 조합)

1) Ji, G. et al., Improvement of the Cathode Electrolyte Interphase on P2-Na_{2/3}Ni_{1/3}Mn_{2/3}O₂ by Atomic Layer Deposition (ACS Appl. Mater. Interfaces, 2017)

2) Sun, L. et al., Recent progress of interface modification of layered oxide cathode material for sodium-ion batteries (Electron (Wiley / HIT), 2024)

3) Kuang, Y. et al., Interface Issues of Layered Transition Metal Oxide Cathodes for Sodium-Ion Batteries: Current Status, Recent Advances, Strategies, and Prospects (Molecules (MDPI), 2024)

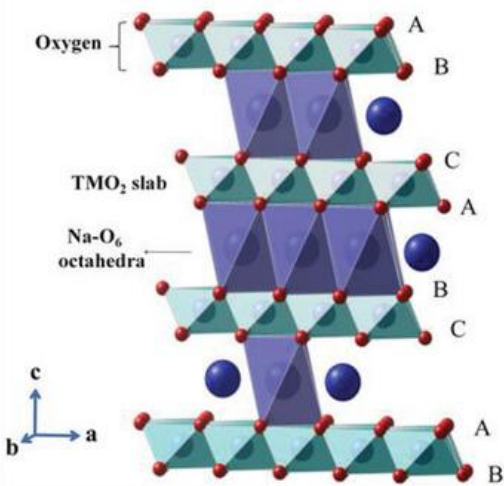




Key 01: 어떤 산화물로 처리를 할까?

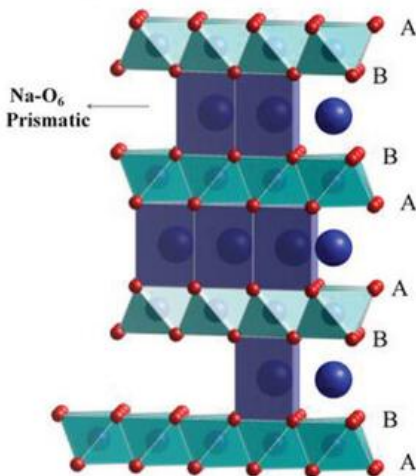
Key 02: 어떤 전구체를 써야 할까?

O3형 [2,3]



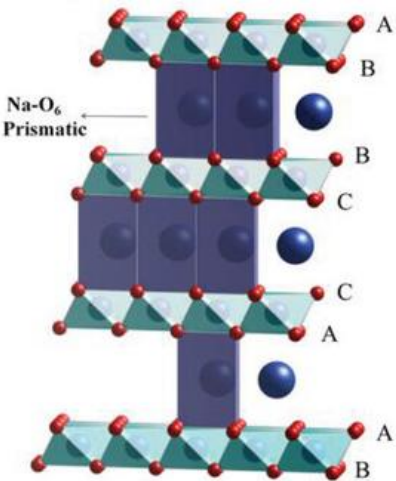
배위: Na: 팔면체(Octahedral)
 적층: ABCABC (3겹 반복)

P2형 [1,2]



배위: Na: 삼각프리즘(Prismatic)
 적층: ABBA (2겹 반복)

P3형 [3]



배위: Na: 삼각프리즘(Prismatic)
 적층: ABBCCA (3겹)



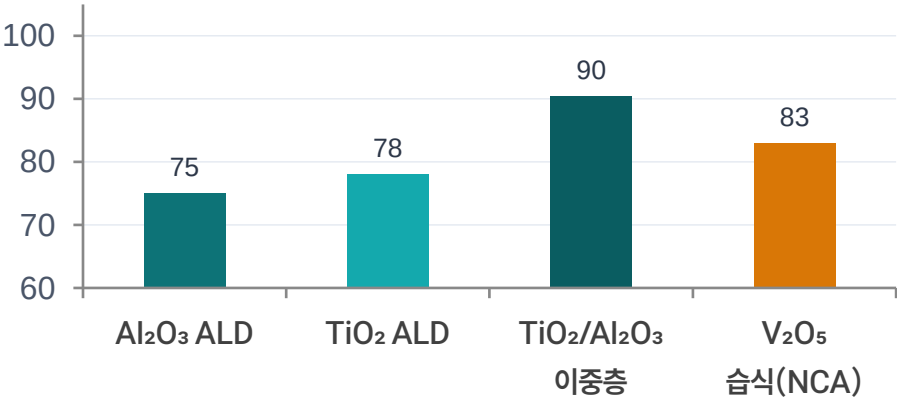


Key 01: 어떤 산화물로 처리를 할까?

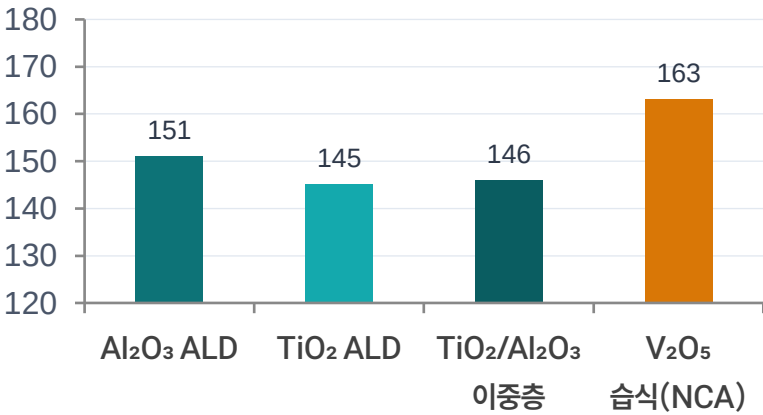
Key 02: 어떤 전구체를 써야 할까?

ALD 코팅 소재별 전기화학 성능 비교

용량 유지율 (%) [1-4]



비용량 (mAh/g) [1-4]





Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

ALD 코팅 소재별 전기화학 성능 비교

코팅 소재	양극재	비용량 (mAh/g)	유지율 (%)	사이클	전지 구성	논문 제목 / 저널
Al ₂ O ₃ ALD	O3-Na[Ni _{0.6} Co _{0.2} Mn _{0.2}]O ₂ (SIB)	151	75	300cy	파우치형 Full Cell (Hard carbon 음극)	[1] Hwang et al. "Resolving the degradation pathways of the O3-type layered oxide cathode surface through the nano-scale aluminum oxide coating..." J. Mater. Chem. A (2017)
TiO ₂ ALD	P2-Na ₂ /3Ni ₁ /3Mn ₂ /3O ₂ (SIB)	~145	78	100cy	Na Half Cell (Na 금속 음극)	[2] Ji et al. "Improvement of the Cathode Electrolyte Interphase on P2-Na ₂ /3Ni ₁ /3Mn ₂ /3O ₂ by Atomic Layer Deposition" ACS Appl. Mater. Interfaces (2017)
TiO ₂ /Al ₂ O ₃ 이중층 ALD	Li-rich LLO Li _{1.08} Ni _{0.34} Co _{0.08} Mn _{0.5} O ₂ (LIB)	146	90.4	100cy @ 1C	Li Half Cell (Li 금속, 2.2-4.6 V)	[3] Chen et al. "Advanced TiO ₂ /Al ₂ O ₃ Bilayer ALD Coatings for Improved Lithium-Rich Layered Oxide Electrodes" ACS Appl. Mater. Interfaces (2024)
V ₂ O ₅ 습식 코팅	LiNi _{0.84} Co _{0.11} Mn _{0.05} O ₂ (LIB, Ni-rich NCA)	163	83 (88.4%/200cy)	200cy @ 1C	Li Half Cell (Li 금속, 2.8-4.3 V)	[4] Wang et al. "V ₂ O ₅ particles anchored on rGO sheets as a dual coating layer for high-nickel cathode material" Appl. Surf. Sci. (2022)
V ₂ O ₅ ALD (SIB 적용)	High-Ni SIB 양극재 (연구공백)	미확인	미확인	—	미실시	해당 없음 (본 연구의 목표)

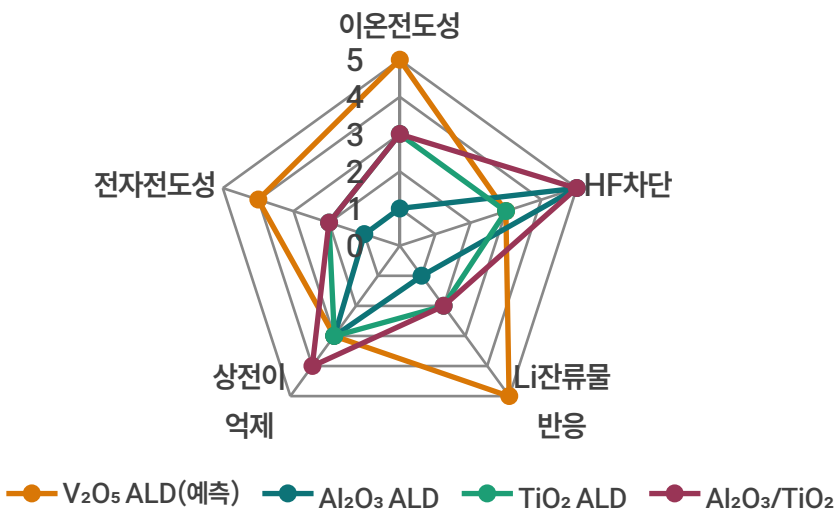




Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

V₂O₅ 코팅 소재별 전기화학 성능 비교



메커니즘	Al ₂ O ₃	TiO ₂	Al ₂ O ₃ /TiO ₂	V ₂ O ₅ (예측)	출처
HF 차단	●●●●●	●●●●○	●●●●●	●●●●○	[1][2]
Li잔류물 제거	●○○○○	●○○○○	●○○○○	●●●●●	[4][7]
이온 전도성	●○○○○	●●●●○	●●●●○	●●●●●	[5][8]
전자 전도성	●○○○○	●●○○○	●●○○○	●●●●○	[4][8]
상전이 억제	●●●●○	●●●●○	●●●●○	●●●●○	[1][3]
산소방출 억제	●●●●○	●●●●○	●●●●○	●●●●●	[7]
코팅층 안정성	●●●●●	●●●●○	●●●●●	●●●●○	[8]





Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

V₂O₅/TiO₂ 이중층 ??

전해질	
TiO ₂ ALD (1-2 nm)	HF 차단 · Rct 최소화 · V ₂ O ₅ 활성 억제 [10]
V ₂ O ₅ ALD (1 nm)	Li/Na 잔류물 제거 · 산소방출 억제 · MIEC [7][8]
High-Ni SIB 양극재	Na[Ni _{0.6} Co _{0.2} Mn _{0.2}]O ₂ 등
❌ 비권장: TiO ₂ (내층)/V ₂ O ₅ (외층) → V ₂ O ₅ 이 전해질에 직접 노출 → 전기화학 활성화 및 용해 위험	

항목	V ₂ O ₅ /Al ₂ O ₃	V ₂ O ₅ /TiO ₂ ✓
HF 차단	★★★★★	★★★★☆
계면 저항	증가 위험	최소화 가능
이종접합	약함	강함 [10]
Na ⁺ 확산	제한	양호
SIB 선례	없음	없음 (공백)

[10] Kim, D. et al., Exploring the Electrochemical Superiority of V₂O₅/TiO₂@Ti₃C₂ MXene Hybrid Nanostructures for Enhanced Lithium-Ion Battery Performance (ACS Appl. Mater. Interfaces, 2024)

*MIEC(Mixed Ionic-Electronic Conductor, 혼합 이온-전자 전도체)는 이온과 전자를 동시에 전달할 수 있는 특수한 소재



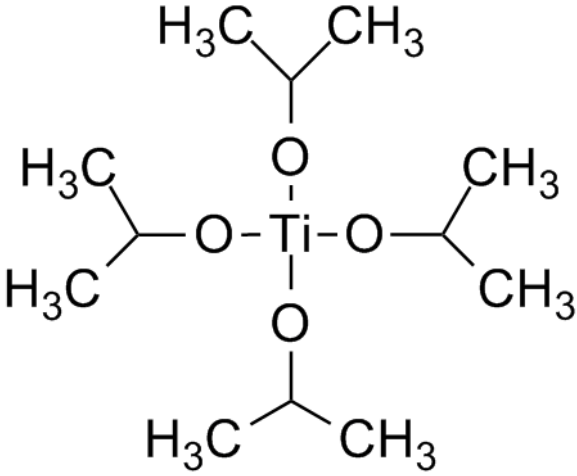


Key 01: 어떤 산화물로 처리를 할까?

Key 02: 어떤 전구체를 써야 할까?

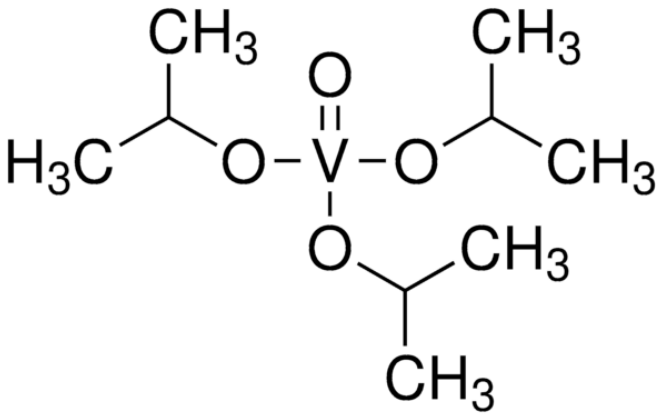
TiO₂ ALD

Precursor A	titanium isopropoxide (Ti(OCH(CH3)2)4) CAS #:546-68-9
Precursor B	H2O



V₂O₅ ALD

Precursor A	Vanadium(V) oxytriisopropoxide (also called vanadyl isopropoxide) OV(OCH(CH3)2)3 CAS #:5588-84-1
Precursor B	H2O





Key 04: 각 전구체의 가열 온도 ? 챔버의 온도는?

Key 05: 증착이 이루어 질까? 결정화도는? 코팅의 안정성은?

TiO₂ ALD

Precursor A	titanium isopropoxide (Ti(OCH(CH3)2)4) CAS #:546-68-9
Precursor B	H2O
Temp. of Precursor A	?
Pulse (Precursor A)	?
Exposure (Precursor A)	?
Purge (Precursor A)	?
Pulse (Precursor B)	?
Exposure (Precursor B)	?
Purge (Precursor B)	?
Chamber Temp	?
Growth Rate	

V₂O₅ ALD

Precursor A	Vanadium(V) oxytriisopropoxide (also called vanadyl isopropoxide) OV(OCH(CH3)2)3 CAS #:5588-84-1
Precursor B	H2O
Temp. of Precursor A	?
Pulse (Precursor A)	?
Exposure (Precursor A)	?
Purge (Precursor A)	?
Pulse (Precursor B)	?
Exposure (Precursor B)	?
Purge (Precursor B)	?
Chamber Temp	?
Growth Rate	





Key 06: ALD 처리가 된 양극재의 안정성이 확보될 수 있을까? 나트륨전지가 잘 구동될까?

Key 07: 시간분할 ALD로 확보된 레서피를 공간분할 ALD에 어떻게 적용해야 할까?

시간분할 ALD Temporal (Chamber)	공간분할 ALD Powall형 (Pneumatic)
전구체 공급 방식	
펄스시간 0.02~5초 정지 기판에 순차 노출	주입속도 10~40 m/s 연속 흐름, 입자 이송
공정 온도	
전형적 80~250°C ALD window 내 유지	전형적 100~250°C 낮을수록 균일 분포
전구체 분리 방식	
퍼지시간 0.05~90초 온도별 최적화 필요	불활성 가스 커튼 구역 간 혼합 차단
처리량 결정 인자	
GPC × 사이클 수 배치당 처리량 제한	이송속도 × 단면적 최대 1000배 확장 보고





Key 06: ALD 처리가 된 양극재의 안정성이 확보될 수 있을까? 나트륨전지가 잘 구동될까?

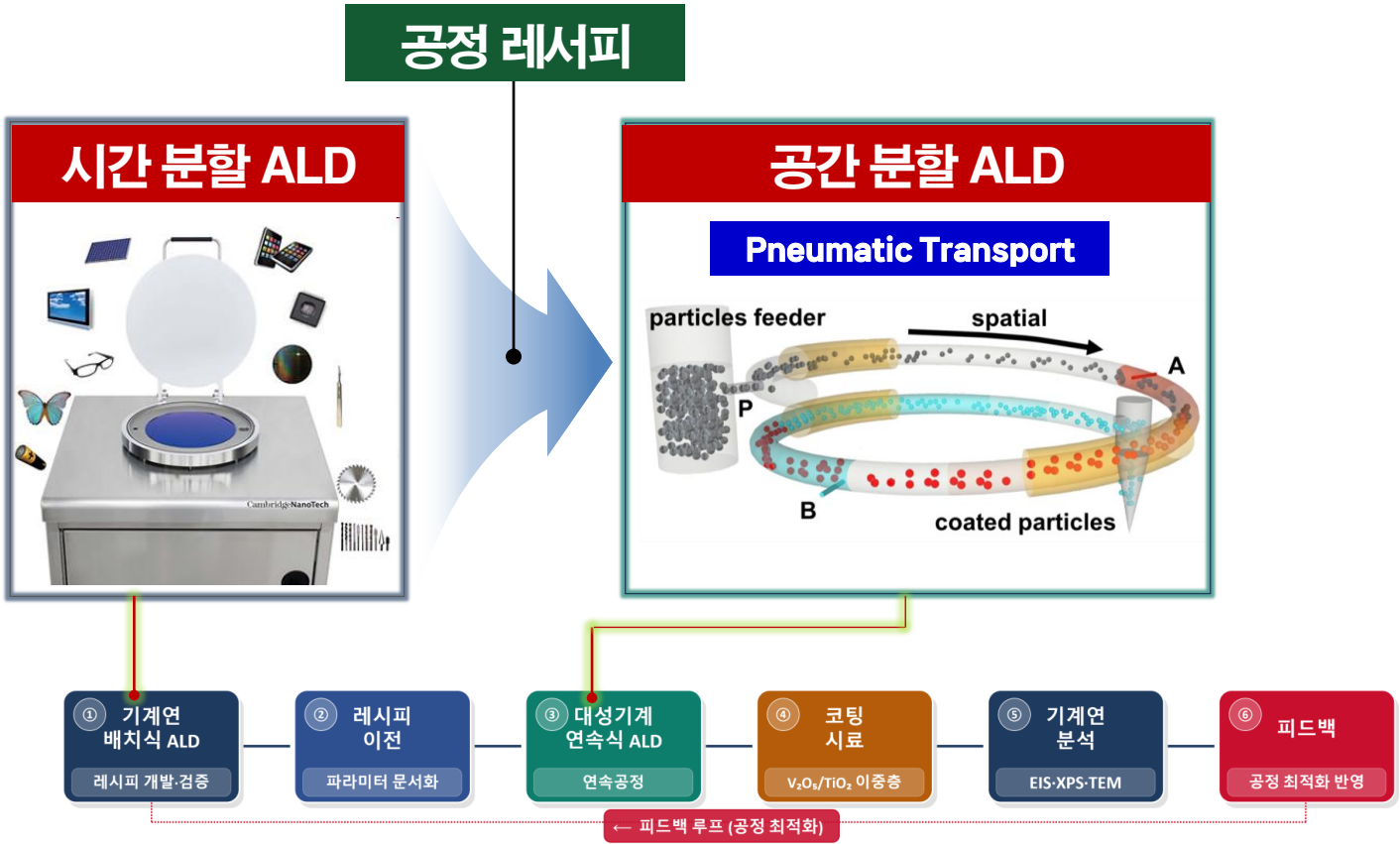
Key 07: 시간분할 ALD로 확보된 레서피를 공간분할 ALD에 어떻게 적용해야 할까?

항목	시간분할 ALD (Temporal)	공간분할 ALD, Powall형 (Spatial)
전구체 분리 원리	펄스-퍼지 순차 노출	반응구역 동시 병행, 입자가 구역 이동
핵심 속도 변수	펄스시간 0.02-5초	전구체 주입속도 10-40 m/s (조절 시 전구체·분말 접촉 향상)
분리/차단 방식	퍼지시간 0.05-90초, 80°C에서 60초 → 200°C에서 12초로 온도에 반비례	반응구역 사이 불활성 가스 커튼으로 격리
온도 최적화 방향	온도 상승 시 반응속도 ↑, 퍼지시간 ↓ 필요	250°C → 100°C로 낮출수록 반응속도 대비 확산속도가 상대적으로 빨라져 나노클러스터 분포가 더 균일해짐
자기제한 조건	펄스시간·퍼지시간 모두 포화(saturation)를 만족해야 자기제한적 성장 보장	주입속도·체류시간이 표면 포화를 만족해야 함 (원리는 동일, 변수만 다름)
처리량 결정식	$GPC \times \text{사이클 수 (배치당 처리량 제한)}$	$\text{이송속도} \times \text{단면적 (연속 공급)}$
처리량 기준값	반응기 부피에 따라 상이	원천기술 기준 약 1 g/min에서 출발, 상업화 과정에서 최소 3자릿수 (1,000배 이상) 증가





Key 06: ALD 처리가 된 양극재의 안정성이 확보될 수 있을까? 나트륨전지가 잘 구동될까?
Key 07: 시간분할 ALD로 확보된 레서피를 공간분할 ALD에 어떻게 적용해야 할까?







- ALD 개요
- 시간 분할 vs 공간 분할 ALD
- 산화물 선택 과정(V_2O_5 / TiO_2 이중층 구조)
- 향후 계획(공정 최적화, 레서피 이전, 전기화학분석)



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수요기관

에코프로비엠, 엘앤에프, 포스코퓨처엠, 엔켐, 코스모신소재



#	저자	논문 제목 (Full Article Title)	저널	권/호	연도	DOI
[1]	Hwang, J.-K. et al.	Resolving the degradation pathways of the O3-type layered oxide cathode surface through the nano-scale aluminum oxide coating for high-energy density sodium-ion batteries	J. Mater. Chem. A	5	2017	10.1039/C7TA08443A
[2]	Ji, G. et al.	Improvement of the Cathode Electrolyte Interphase on P2-Na ₂ /3Ni ₁ /3Mn ₂ /3O ₂ by Atomic Layer Deposition	ACS Appl. Mater. Interfaces	9	2017	10.1021/acsaami.7b05326
[3]	Chen, W.-M. et al.	Advanced TiO ₂ /Al ₂ O ₃ Bilayer ALD Coatings for Improved Lithium-Rich Layered Oxide Electrodes	ACS Appl. Mater. Interfaces	16	2024	10.1021/acsaami.3c16948
[4]	Wang, L. et al.	A new modification strategy for improving the electrochemical performance of high-nickel cathode material: V ₂ O ₅ particles anchored on rGO sheets as a dual coating layer	Appl. Surf. Sci.	579	2022	10.1016/j.apsusc.2021.152037
[5]	Li, L. et al.	Effect of TiO ₂ Coating on Structure and Electrochemical Performance of LiNi _{0.6} Co _{0.2} Mn _{0.2} O ₂ Cathode Material for Lithium-Ion Batteries	Materials (MDPI)	17	2024	10.3390/ma17246222
[6]	CATL	CATL Naxtra Na-ion Battery Technology (EV, 2024); CATL BESS Na-ion Cell — 300 Ah+, 160 Wh/kg, 15,000 cycles	CATL Technology Announcement / ESI E	—	2024/2026	catl.com/news
[7]	Mao, G. et al.	Improving the electrochemical performance of LiNi _{0.8} Co _{0.15} Al _{0.05} O ₂ cathode material by V ₂ O ₅ coating	J. Alloys Compd.	892	2022	10.1016/j.jallcom.2021.162161
[8]	Chen, X. et al.	MWCNT/V ₂ O ₅ Core/Shell Sponge for High Areal Capacity and Power Density Li-ion Cathodes; and ALD V ₂ O ₅ from VO(thd) ₂ /O ₃ : crystal thin film cathodes with 330 mAh/g	ACS Nano / Chem. Mater.	6/24	2012	10.1021/nn204451r
[9]	O'Donoghue, J. F. Rohan et al.	Nanoengineering of Thin Film V ₂ O ₅ Cathode Interfaces via Atomic Layer Deposition for use with Polymer Gel Ionic Liquid Electrolytes	Batteries & Supercaps	7	2024	10.1002/batt.202300447
[10]	Kim, D. et al.	Exploring the Electrochemical Superiority of V ₂ O ₅ /TiO ₂ @Ti ₃ C ₂ MXene Hybrid Nanostructures for Enhanced Lithium-Ion Battery Performance	ACS Appl. Mater. Interfaces	16	2024	10.1021/acsaami.4c10656
[11]	Liu, J. et al.	Enhancing V ₂ O ₅ Cathode Performance through Heterostructure: Reduced Polar-Lattice Interaction	J. Phys. Chem. C	128	2024	10.1021/acs.jpcc.3c08078
[12]	Sun, L. et al.	Recent progress of interface modification of layered oxide cathode material for sodium-ion batteries	Electron (Wiley / HIT)	2	2024	10.1002/elt.231
[13]	Kuang, Y. et al.	Interface Issues of Layered Transition Metal Oxide Cathodes for Sodium-Ion Batteries: Current Status, Recent Advances, Strategies, and Prospects	Molecules (MDPI)	29	2024	10.3390/molecules29245988
[14]	Xi, X. et al.	Enhanced cyclic stability of NCM-622 cathode by Ti ³⁺ doped TiO ₂ coating	J. Alloys Compd.	868	2021	10.1016/j.jallcom.2021.159183
[15]	Huang, K. et al.	TiO ₂ -LiF composite coating for improving NCM622 cathode cycling stability: one-step construction	RSC Advances	13	2023	10.1039/d3ra05659g

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